物理化学（3）

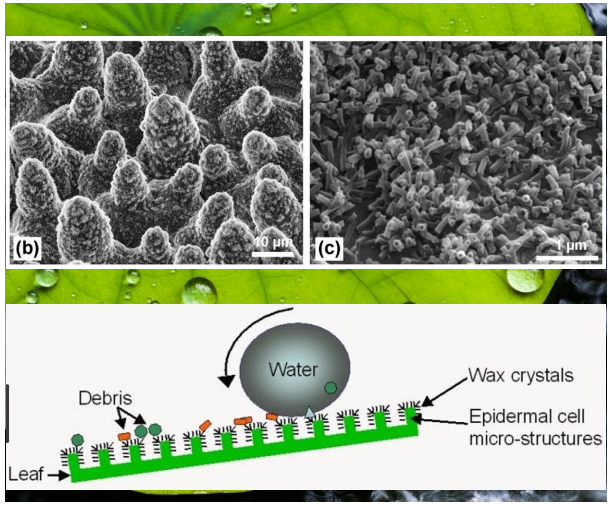
表面化学



清華大學
Tsinghua University

Surface Phenomena

lotus leaf effect



dancing droplets



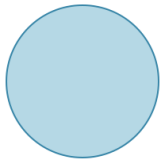
lying droplets



Surface science (表面科学) is the study of physical and chemical phenomena that occur at the interface of two phases. The effects of ***surface tension*** (表面张力) are of central importance in many everyday phenomena.

Surface Science at Nanoscale

The surface-size dependence

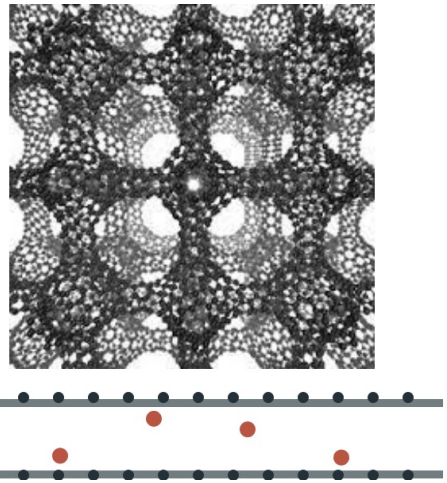


Particle size	10 cm	1 mm	1 μm	1 nm
Surface	1	100	100,000	100,000,000

$$\frac{A_{\text{tot}}}{V_{\text{tot}}} = \frac{6}{d}, \qquad \frac{A_{\text{tot}}}{A_0} = \frac{d_0}{d}$$

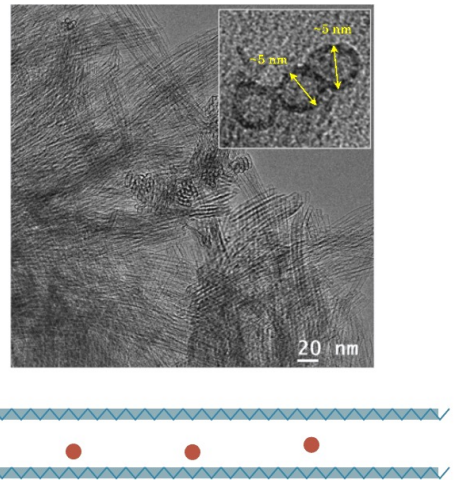
等总体积：粒径越小，总表面积越大

Porous materials



连通孔道提供内部表面分子在孔道中迁移并接近孔壁

Nanotube



内外表面与纳米孔道共同影响材料性质

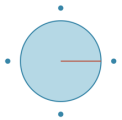
Surface science becomes increasingly more important for understanding materials properties as the material size becomes smaller. At a nano scale, the surfaces can even dominantly determine the materials properties, playing the critical role together with the confinement effects.

Content

surface tension
(表面张力)

liquid-vapor interface

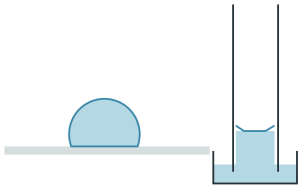
- surface of a pure liquid
  - curved surface
  - Young-Laplace equation
  - Kelvin equation
  - size-dependent equilibrium
  - metastable phase



$$\Delta p = 2\gamma / R$$

- surface of a solution
  - surface adsorption
  - Gibbs surface excess
  - surfactant
  - applications of surfactant

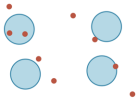
solid-liquid interface



$$\gamma_{SV} - \gamma_{SL} = \gamma_{LV} \cos\theta$$

- wetting and spreading
- capillary effect
- methods for measuring surface tension

- **colloidal systems**
- **emulsion systems**



gas-solid interface

- adsorption theories
  - Langmuir theory
  - BET theory

$$\Theta = Kp/(1+Kp)$$

- heterogeneous catalysis
  - unimolecular
  - Langmuir-Hinshelwood mechanism
  - Eley-Rideal mechanism

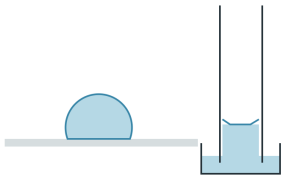
Aim

- You should ***understand*** the concept of surface tension and know how to use it to explain surface/interface phenomena.
- You should ***know*** how to use the Young-Laplace equation, the Kelvin equation and the Gibbs adsorption excess to calculate equilibrium problems.
- You should ***understand*** the capillary effect and be able to judge whether a liquid wet or spread on a solid surface.
- You should ***know*** the methods to determine surface tension.
- You should ***know*** the definition of surfactants and their properties.
- You should ***master*** the Langmuir theory and the BET theory and can ***use*** them to calculate surface area and adsorption enthalpy.
- You should ***understand*** the mechanism of heterogenous catalysis.
- You should ***know*** the definition and properties of colloids.
- You should ***know*** the definition and properties of emulsions.

Teaching Objectives

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界面与润湿



$$\Delta p = 2\gamma / R$$

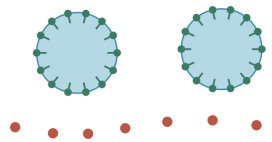
吸附与催化





$$\Theta = Kp/(1+Kp)$$

胶体与乳液

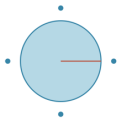


Content

surface tension
(表面张力)

liquid-vapor interface

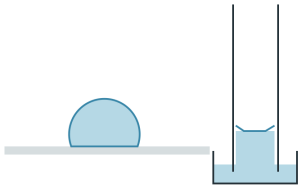
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  - Gibbs surface excess
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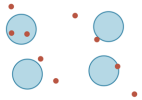
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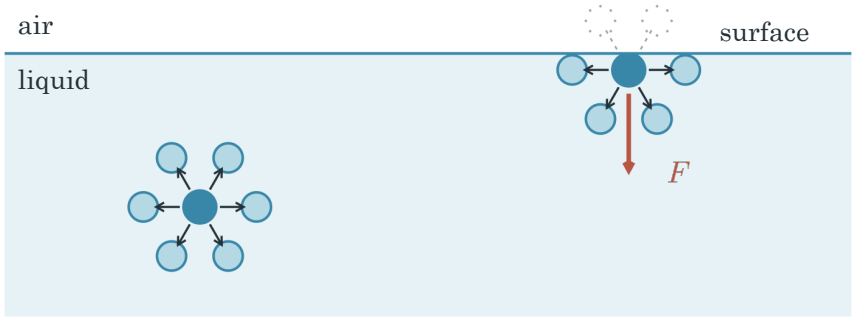
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- heterogeneous catalysis
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  - Langmuir-Hinshelwood mechanism
  - Eley-Rideal mechanism

Surface Tension

- The molecules at a surface are different from those in the bulk phase. Thus, the inhomogeneity at the surface results in a force whose direction goes towards the bulk phase, and this force is called ***surface tension*** (表面张力).

- Take liquid–air interfaces for example, surface tension results from the greater attraction of liquid molecules to each other (due to *cohesion*, 凝聚力) than to the molecules in the air (due to *adhesion*, 粘附力).



- We need to apply work to move molecules in the bulk phase to the surface. We thus define *surface work* (表面功) as the work needed to ***reversibly*** increase the surface area at given temperature, pressure and composition. As such, we can define the following constant of proportionality as surface tension:

$$\delta W' = -\gamma\, dA \quad \Rightarrow \quad W' = -\int_{A_1}^{A_2} \gamma\, dA$$

$$W' = -\Delta G = -\gamma A$$



surface expansion

Surface energy (表面能) equals to γA

(Note that surface energy is not the total free energy of surface molecules, but is the excess energy compared with equivalent molecules in the bulk!)

Surface tension equals to *surface energy per unit area* (比表面能), so the unit is $\mathrm{J \cdot m^{-2}}$.

Surface Tension

- As explained before, surface tension is a kind of force that tends to shrink the surface into its minimum. We define *the force acting on per unit length* ($\mathrm{N \cdot m^{-1}}$) as surface tension, and let's use γ' to represent this force.

- Assume that we *reversibly* lift a thin film from the liquid phase, whose width is l and height is a very small number dx. Then let's calculate the work we need to apply:

$$\delta W' = -F\,dx \qquad F = \gamma' \cdot 2l$$

$$\delta W' = -\gamma' \cdot 2l\,dx = -\gamma'\,dA$$

$$\gamma' = \gamma$$

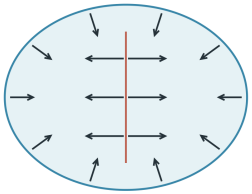
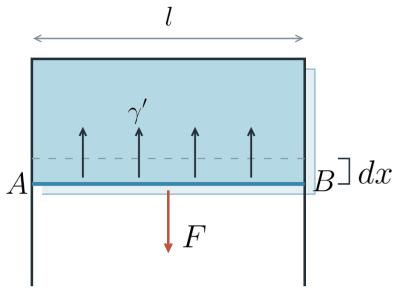


图 11-3　表面张力的方向



双面液膜

图 11-4　表面张力与比表面能的关系推导

Surface tension has two different interpretations and two different units. But they have the same value and are essentially the same.

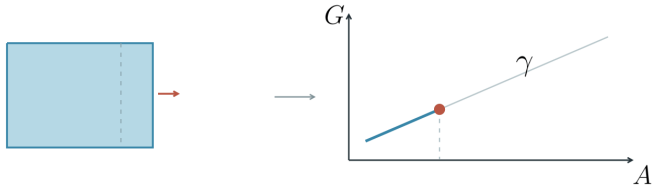
表面张力有两种不同的解释（比表面能和表面张力），因此有两种不同的单位(J $\mathrm{m^{-2}}$ 和 N $\mathrm{m^{-1}}$)但它们本质上是相同的，都是表面不均匀力场的宏观量度。

Surface Tension

According to the previous definition, surface tension equals to the work needed to reversibly increase the per unit surface area at given temperature, pressure and composition. Consequently, we can derive another expression of surface tension as:

$$\gamma = \left(\frac{\partial G}{\partial A}\right)_{T,P,n_B,n_C,\ldots}$$

固定温度、压力和各组分物质的量



If we take surfaces into consideration, the Gibbs free energy of a system should be written as:

$$G = f(T, p, n_B, n_C, \cdots, A)$$

Thus we have:

$$dG = -S\,dT + V\,dp + \sum_B \mu_B\,dn_B + \gamma\,dA$$

Similarly:

$$dU = T\,dS - p\,dV + \sum_B \mu_B\,dn_B + \gamma\,dA$$

$$dH = T\,dS + V\,dp + \sum_B \mu_B\,dn_B + \gamma\,dA$$

$$d(A) = -S\,dT - p\,dV + \sum_B \mu_B\,dn_B + \gamma\,dA$$

Thus we have other expressions of γ

$$\gamma = \left(\frac{\partial U}{\partial A}\right)_{S,V,n_B,n_C,\ldots}$$

$$\gamma = \left(\frac{\partial H}{\partial A}\right)_{S,p,n_B,n_C,\ldots}$$

$$\gamma = \left(\frac{\partial (A)}{\partial A}\right)_{T,V,n_B,n_C,\ldots}$$

Effects of Temperature and Pressure

$$dG = -S\,dT + V\,dp + \sum_B \mu_B\,dn_B + \gamma\,dA$$

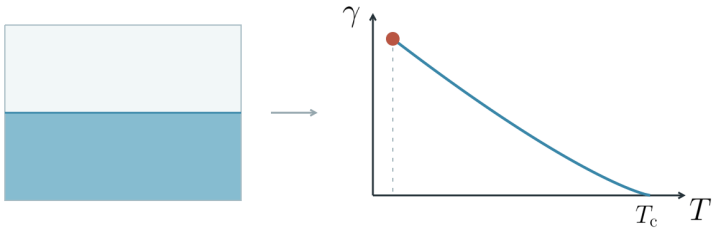
$$\left(\frac{\partial\gamma}{\partial T}\right)_{A,P,n_B,n_C,\ldots} = -\left(\frac{\partial S}{\partial A}\right)_{T,p,n_B,n_C,\ldots} < 0$$

Surface tension decreases with temperature.

When temperature increases, the inhomogeneity at the surface becomes weakened, which results in a smaller surface tension.

At the critical point (and above), the surface tension completely disappears.

升温时，两相差异减小，界面逐渐消失



$$dG = -S\,dT + V\,dp + \sum_B \mu_B\,dn_B + \gamma\,dA$$

$$\left(\frac{\partial\gamma}{\partial p}\right)_{T,A,n_B,n_C,\ldots} = \left(\frac{\partial V}{\partial A}\right)_{T,p,n_B,n_C,\ldots} \approx 0$$

- The influence of pressure on surface tension is small and usually can be neglectable.



表 11-1 $293.15\ \mathrm{K}$ **时的表面张力**

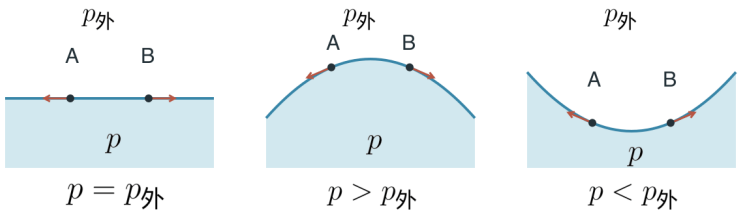
物质	$\gamma/(\mathrm{N\cdot m^{-1}})$	物质	$\gamma/(\mathrm{N\cdot m^{-1}})$
水	0.07288	辛烷	0.02162
苯	0.02888	庚烷	0.02014
甲苯	0.02852	丙酸	0.02669
甲醇	0.02250	丁酸	0.02651
乙醇	0.02239	硝基甲烷	0.03266

Curved Surfaces and Young-Laplace Equation

- The minimization of the surface area of a liquid commonly results in the formation of a curved surface (弯曲表面): A bubble (气泡) is a region in which vapor (and possibly air too) is trapped by a thin film; A droplet (液滴) is a small volume of liquid surrounded by its vapor (and possibly also air).

- The tendency for a cavity to minimize its surface area results in the pressure inside the cavity (内压) being greater than that outside (assuming the concave side is liquid). This pressure difference is called 附加压力 in Chinese, and it is radius dependent.

平面、凸面与凹面的压差

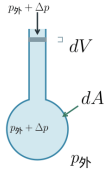


- Step 1: The work applied to enlarge the droplet equals to

$$\delta W = -(p_{outside} + \Delta p)\, dV + p_{outside} dV$$

$$\downarrow$$

$$\delta W = -\Delta p \cdot dV$$



- Step 2: The work needed to increase the surface area equals to

$$\delta W = -\gamma \cdot dA$$

$$\downarrow$$

$$-\Delta p \cdot dV = -\gamma \cdot dA \quad \rightarrow \quad \Delta p = \gamma \cdot \frac{dA}{dV}$$

- Step 3: Calculate dA and dV.

$$dA = d(4\pi r^2) = 8\pi r\, dr \qquad dV = d\left(\frac{4}{3}\pi r^3\right) = 4\pi r^2 dr$$

$$\downarrow$$

$$\Delta p = \frac{2\gamma}{r}$$

Young-Laplace equation

Curved Surfaces and Young-Laplace Equation

Some interpretations:

- The pressure difference is proportional to surface tension.
- The pressure difference is inversely proportional to radius.
- The sign of radius:
  - for convex meniscus (凸液面): $r > 0$
  - for concave meniscus (液滴): $r < 0$
- For a bulb filled with air, the pressure difference inside and outside can be calculated as:

$$\Delta p = \frac{4\gamma}{r}$$

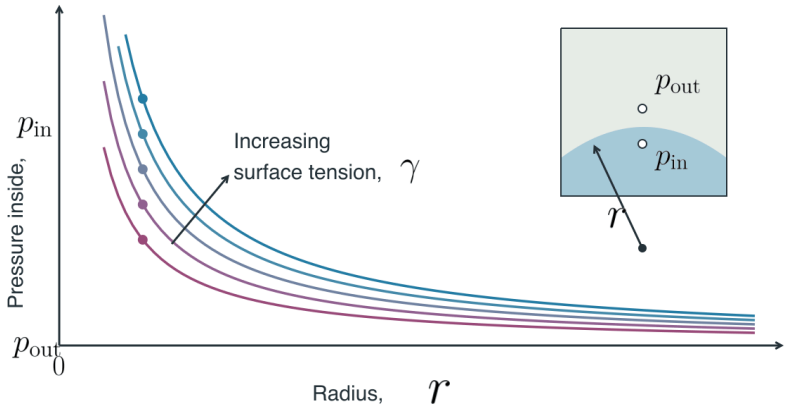
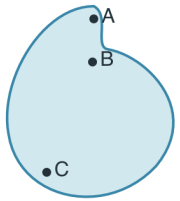


Figure 14C.5 The dependence of the pressure inside a curved surface on the radius of the surface, for increasing values of the surface tension.

非球形界面趋于圆整（示意）



Kelvin Equation

A review of phase equilibrium:

At liquid-vapor equilibrium:

$$\mu(\mathrm{l}) = \mu(\mathrm{g}) \qquad d\mu(\mathrm{l}) = d\mu(\mathrm{g})$$

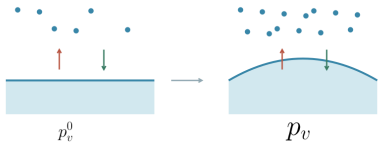
At isothermal condition:

$$V_m(\mathrm{l})\, dp = V_m(\mathrm{l})\, dp_v$$

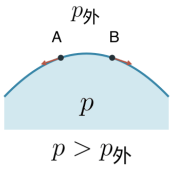
$$\left(\frac{\partial p_v}{\partial p}\right)_T = \frac{V_m(\mathrm{l})}{V_m(\mathrm{g})}$$

The equilibrium vapor pressure of a liquid is pressure dependent: The higher the pressure of the liquid, the higher of equilibrium vapor pressure it has.

For a liquid with curved surfaces, what would it happen to its equilibrium vapor pressure?



- **Step 1: Since** $(p + \Delta p) > p$, the chemical potential inside the droplet is higher:



$$\Delta\mu = \int_p^{p+\Delta p} V_m(\mathrm{l})\, dp$$

$$\Delta\mu = V_m(\mathrm{l}) \cdot (p + \Delta p - p) = V_m(\mathrm{l}) \cdot \Delta p = \frac{M}{\rho} \cdot \Delta p$$

- **Step 2: The Young-Laplace equation:** $\Delta\mu = \dfrac{M}{\rho} \cdot \dfrac{2\gamma}{r}$

- **Step 3: Consider the vapor phase:**
  Use p_v and p_v^0 to represent the vapor pressure for the liquid with curved surface and flight surface. Thus:

$$\Delta\mu = \int_p^{p+\Delta p} V_m(\mathrm{g})\, dp = RT \ln \frac{p_v}{p_v^0}$$

$$\ln \frac{p_v}{p_v^0} = \frac{2M\gamma}{RT\rho r}$$

Kelvin equation

Kelvin Equation

Some interpretations:

- The change of vapor pressure for a curved surface can be interpreted by the unbalanced force on surface molecules.
  - for convex meniscus (凸液面): surface molecules have a higher tendency to leave the liquid phase to the vapor phase.
  - for concave meniscus (凹液面): surface molecules have a lower tendency to leave the liquid phase to the vapor phase.
- The change of vapor pressure is much apparent when the droplets are nanosized.
- The Kelvin equation is applicable to a solid-vapor surface.

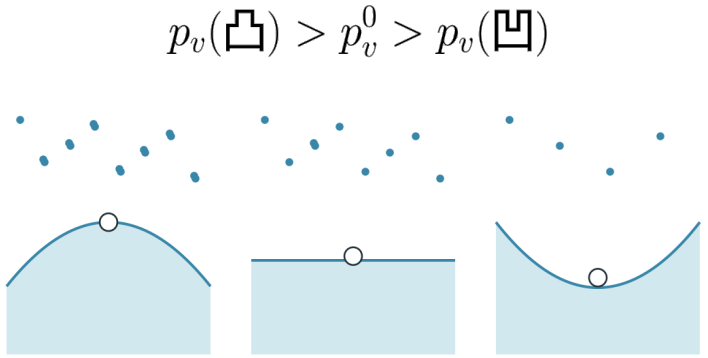


图 11-9　不同液面蒸气压的相互比较

表 11-2　$293.15\,\mathrm{K}$ **时水的蒸气压与水滴半径的关系**

r/m	10^{-9}	10^{-8}	10^{-7}	10^{-6}
p_v/p_v^0	2.95	1.114	1.011	1.001

The Effect of Particle Size on Solubility

A review of phase equilibrium:

For a solute at equilibrium with its solid phase:

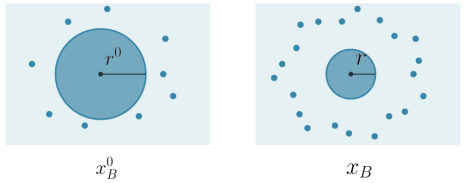
$$\mu_B(\mathrm{s}) = \mu_B(\mathrm{sln})$$

At isothermal condition:

$$\left(\frac{\partial \mu_B(\mathrm{s})}{\partial p}\right)_T = V_m(\mathrm{s}) > 0$$

The chemical potential of a solid increases with pressure. According to the Young-Laplace equation, we know that the chemical potential of a solid particle increases when its size decreases.

Then, we can imagine that the solubility of that solid particle increases accordingly.



- At phase equilibrium, we have:

$$B(\mathrm{s}, r, T, p+\Delta p) \underset{\leftrightarrow}{\text{equilibrium}} B(\mathrm{sln}, T, p, x_B)$$

$$\mu_B(\mathrm{s}, T, p+\Delta p) = \mu_B(\mathrm{sln}, T, p, x_B) \qquad d\mu_B(\mathrm{s}) = d\mu_B(\mathrm{sln})$$

$$d\mu_B(\mathrm{s}) = -S_m(\mathrm{s})\, dT + V_m(\mathrm{s})\, d(p+\Delta p)$$

$$d\mu_B(\mathrm{sln}) = -S_m(\mathrm{sln})\, dT + V_m(\mathrm{sln})\, dp + RT d\ln x_B$$

- Assuming T and p are fixed:

$$V_m(\mathrm{s})\, d(\Delta p) = V_m(\mathrm{s})\, d\left(\frac{2\gamma}{r}\right) = RT d\ln x_B$$

- Use x_B and x_B^0 to represent solubility for the solid particle with a radius of r and that of the bulk (r^0).

$$\int_{r^0}^{r} V_m(\mathrm{s})\, d\left(\frac{2\gamma}{r}\right) = \int_{x_B^0}^{x_B} RT d\ln x_B \qquad r^0 \gg r$$

$$\ln\frac{x_B}{x_B^0} = \frac{2\gamma M_B}{RT\rho_B r}$$

The solubility indeed increases when the particle size decreases.

The Effect of Particle Size on Melting Point

- At melting point (T_m), we have:

$$B(\mathrm{s}, r, T_m, p+\Delta p) \underset{\leftrightarrow}{\text{equilibrium}} B(\mathrm{l}, T_m, p)$$

$$\mu_B(\mathrm{s}, T_m, p+\Delta p) = \mu_B(\mathrm{l}, T_m, p) \qquad d\mu_B(\mathrm{s}) = d\mu_B(\mathrm{l})$$

$$d\mu_B(\mathrm{s}) = -S_m(\mathrm{s})\, dT + V_m(\mathrm{s})\, d(p+\Delta p)$$

$$d\mu_B(\mathrm{l}) = -S_m(\mathrm{l})\, dT + V_m(\mathrm{l})\, dp$$

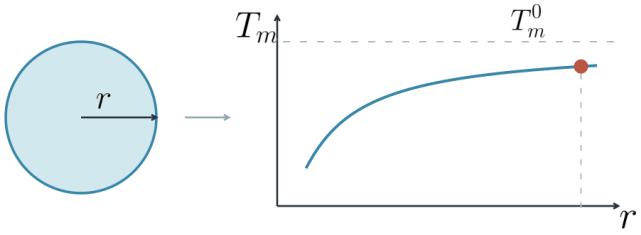
- Assuming p is fixed ($dp = 0$):

$$-S_m(\mathrm{s})\, dT + V_m(\mathrm{s})\, d\left(\frac{2\gamma}{r}\right) = -S_m(\mathrm{l})\, dT$$

$$\Delta_s^l S_m dT + V_m(\mathrm{s})\, d\left(\frac{2\gamma}{r}\right) = 0$$

$\Delta_s^l S_m$ represents the change of molar entropy of melting, which approximately equals to $\frac{\Delta_s^l H_m}{T_m}$.

$$\frac{\Delta_s^l H_m}{T_m} dT + V_m(\mathrm{s})\, d\left(\frac{2\gamma}{r}\right) = 0 \quad \rightarrow \quad \left(\frac{\partial \ln T_m}{\partial r}\right)_p = \frac{2\gamma M}{\Delta_s^l H_m \rho r^2}$$



按公式：粒径减小，熔点降低（原页首条结论有误）

- Use T_m and T_m^0 to represent melting point for the solid particle with a radius of r and that of the bulk (r^0).

$$\int_{T_m^0}^{T_m} d\ln T_m = \frac{2\gamma M}{\Delta_s^l H_m \rho} \int_{r^0}^{r} \frac{1}{r^2} dr$$

$r^0 \gg r$

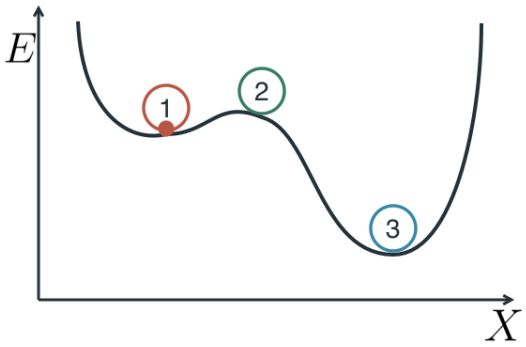
$$\ln \frac{T_m}{T_m^0} = \frac{-2\gamma M}{\Delta_s^l H_m \rho r}$$

$\Delta_s^l H_m > 0$

- The melting point increases when the particle size of the solid decreases.
- The melting point of nanoparticles is much lower that that of bulk particles.

Metastable Phase

- In chemistry and physics, metastability (亚稳态) denotes an intermediate energetic state within a dynamical system other than the system's state of least energy.



A metastable state of weaker bond (1), a transitional 'saddle' configuration (2) and a stable state of stronger bond (3).

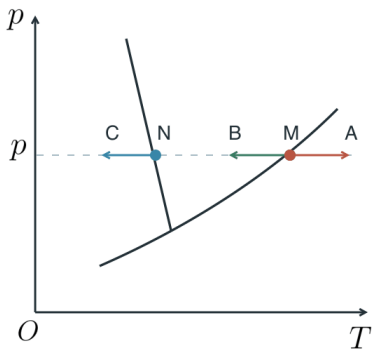


图 11-11　水的亚稳状态

Three metastable phases of water:

- Supersaturated vapor (过饱和蒸气) denotes a metastable state in which water maintains as a vapor phase although the vapor pressure is higher than the equilibrium vapor pressure.
- On the other hand, water may be superheated above its boiling temperature (过热水) and supercooled below its freezing temperature (过冷水).
- *In each case, the thermodynamically stable phase is not achieved on account of the kinetic stabilization that occurs in the absence of nucleation center.*

  To form such nucleation centers, we create new interfaces, and thus surface energy *(* γA *) is needed as additional energy input.*

An Example

例 11-1　在正常沸点时，如果水中仅含有直径为 $10^{-3}\,\mathrm{mm}$ 的空气泡，问使这样的水开始沸腾需过热多少度？水在 $100^{\circ}\mathrm{C}$ 时的表面张力 $\gamma = 0.0589\,\mathrm{N\cdot m^{-1}}$，摩尔汽化热 $\Delta_l^g H_m = 40656\,\mathrm{J\cdot mol^{-1}}$。

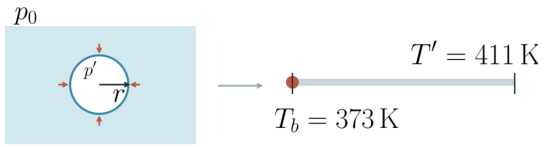
解：含空气泡内的压力为 p'，空气泡外的压力 $p_0 = 101325\,\mathrm{Pa}$，空气泡的半径 $r = 5\times10^{-7}\,\mathrm{m}$

根据 Young-Laplace 方程

$$p' - p_0 = \frac{2\gamma}{r}$$

$$p' = p_0 + \frac{2\gamma}{r} = \left(101325 + \frac{2\times0.0589}{5\times10^{-7}}\right)\mathrm{Pa} = 336925\,\mathrm{Pa}$$

即泡内压力约为大气压力的 3.3 倍。在外压为 p' 时沸腾的温度 T' 可由 Clausius-Clapeyron 方程求算：



$$\ln\frac{p'}{p_0} = -\frac{\Delta_l^g H_m}{R}\left(\frac{1}{T'} - \frac{1}{T_b}\right)$$

$$\ln\frac{336925}{101325} = -\frac{40656}{8.314}\left(\frac{1}{T'/\mathrm{K}} - \frac{1}{373}\right)$$

解得 $T' = 411\,\mathrm{K}$

显然 T' 即为开始沸腾的温度，所以过热的温度为

$$\Delta T = T' - T_b = (411 - 373)\,\mathrm{K} = 38\,\mathrm{K}$$

即这样的水约在 $138^{\circ}\mathrm{C}$ 时开始沸腾。

Metastable Phase

Why is superheated water possible?

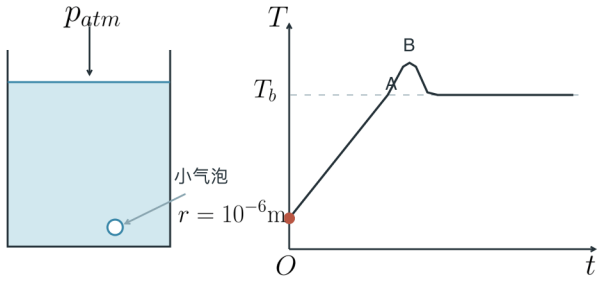


图 11-12　液体的过热现象

图 11-13　纯液体的升温与沸腾过程

- At normal boiling point　　$T_b,\ p_v^0 = p_{atm}$
- Assuming the first bubble appears in the size of 10^{-6}m, $p_v = p_v^0 + \Delta p > p_{atm}$
- The boiling point at　　$p_v = p_v^0 + \Delta p$ would be　T_b', and obviously　$T_b' > T_b$
- For a T in the range of　$T_b' > T > T_b$, the water would be superheated water.

How to avoid superheated state? Add some zeolites.

Why is supercooled water possible?

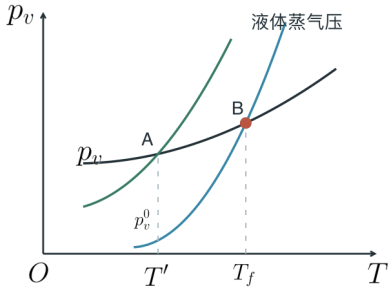
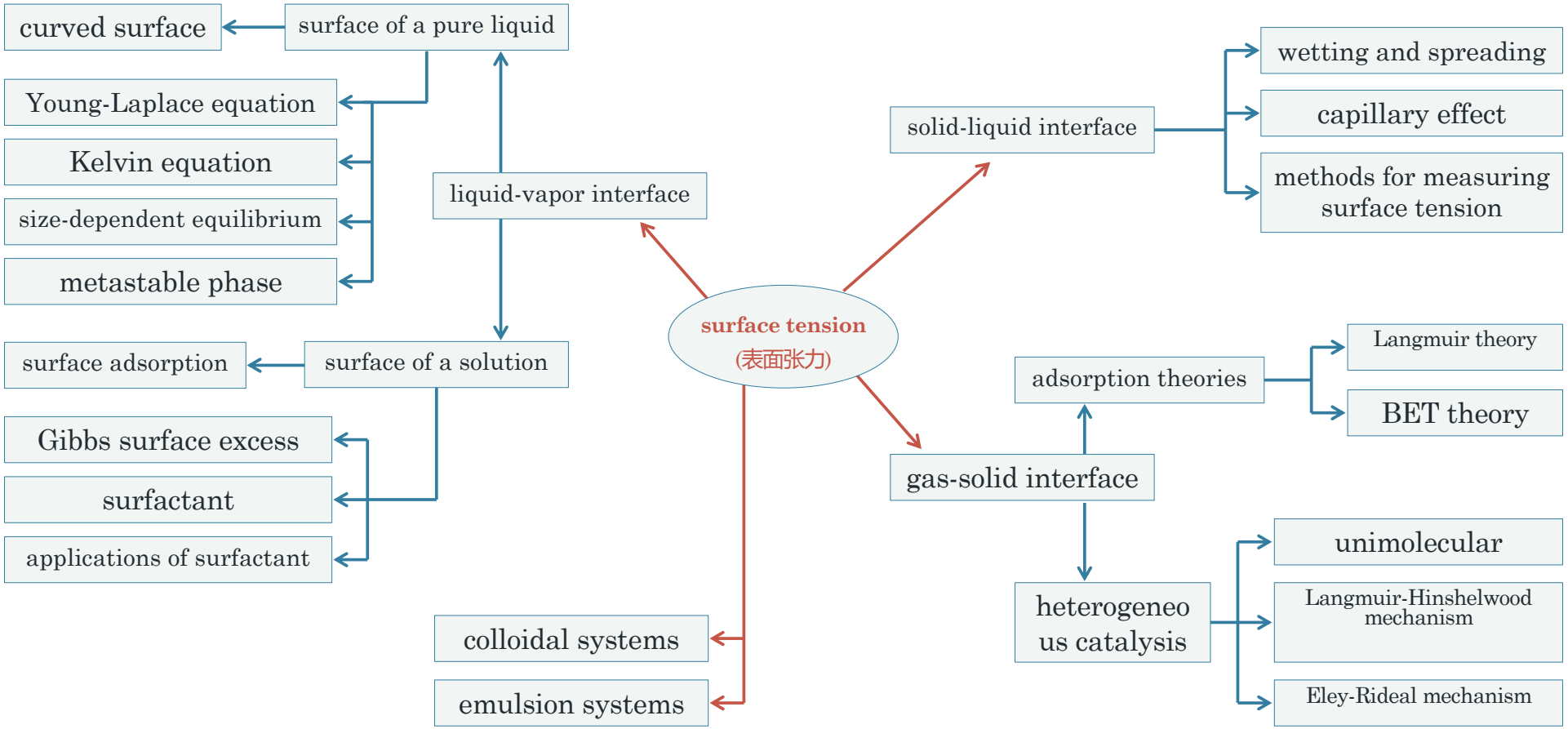


图 11-14　液体过冷现象的说明

- At normal freezing point　　$T_f,\ p_v^0 = p_{atm}$
- Assuming the first ice particle appears in the size of 10^{-6}m, $p_v = p_v^0 + \Delta p > p_{atm}$
- The freezing point at　　$p_v = p_v^0 + \Delta p$ would be　T_f', and obviously　$T_f' < T_f$
- For a T in the range of　$T_f' < T < T_f$, the water would be supercooled water.

How to create supercooled water? Keep it clean!

A Mind Map for Surface Science



Surface of A Solution

The surface energy (γA) of a solution can be minimized in two ways:

- Shrink spontaneously to reduce the surface area.
- Adsorb more solute molecules into its surface to reduce surface tension.

Consequently, the composition of solution surface might be different from that of the bulk due to surface adsorption (表面吸附). We consider the surface as a phase and use the letter of σ to represent the surface phase (表面相)

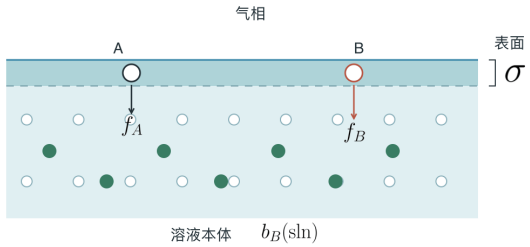


图 11-25 溶液的表面吸附

- Depending on the type of solute, the change of surface tension of water behaves differently:

  Type I: inorganic salts, which slightly increase the surface tension of water.

  Type II: organics, which reduce the surface tension of water, with an approximate limit

  Type III: surfactants, which greatly reduce the surface tension of water even with a low concentration.

- For the third kind of substances, which can greatly reduce the surface tension of a liquid. We call them as surfactant (表面活性剂).

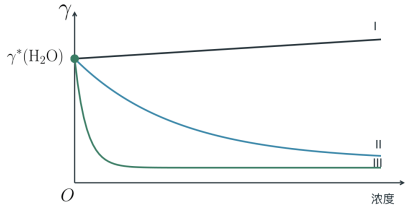


图 11-24 水溶液的表面张力与浓度的关系

Gibbs Surface Excess

- Due to surface adsorption, the concentration of solute (B) in the surface phase $c_B(\sigma)$ is different from that in the bulk phase $c_B(\mathrm{A})$
- At adsorption equilibrium, the composition of surface phase is unchanged: $n_A(\sigma) + n_B(\sigma)$
- For the solvent in the surface phase $n_A(\sigma)$, if it is in the bulk phase, it would dissolve B with an amount of n'_B. We know that $n_B(\sigma) \neq n'_B$ and the difference is $\Delta n_B = n_B(\sigma) - n'_B$

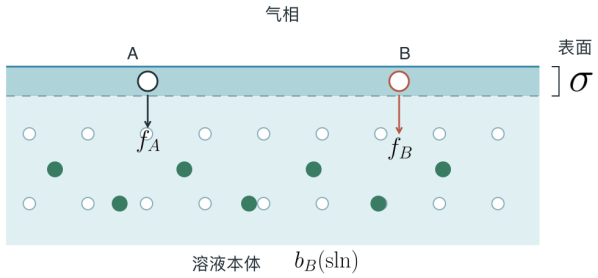
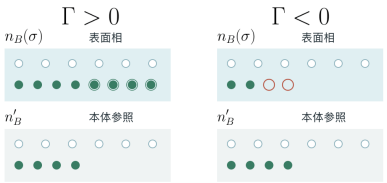


图 11-25　溶液的表面吸附

- To quantitively describe such a difference, we define surface excess (表面超量) as follows:

$$\Gamma = \frac{\Delta n_B}{A}$$ surface excess

- The unit of surface excess: $\mathrm{mol \cdot m^{-2}}$
- Surface excess represents the additional amount of molecules the solvent in the surface phase of $1\mathrm{m}^{-2}$ could dissolve than does the equivalent amount of solvent the bulk phase.
- The sign of surface excess can be minus or plus, depending on the behavior of surface adsorption.



Gibbs Surface Excess

Derive the expression of Gibbs surface excess:

Step 1: The Gibbs free energy of the surface phase

$$G = f(T, p, n_B, n_C, \cdots, A)$$

$$G(\sigma) = f[T, p, n_A(\sigma), n_B(\sigma), A]$$

$$G(\sigma) = n_A(\sigma)\left(\frac{\partial G(\sigma)}{\partial n_A(\sigma)}\right) + n_B(\sigma)\left(\frac{\partial G(\sigma)}{\partial n_B(\sigma)}\right) + A\left(\frac{\partial G(\sigma)}{\partial A}\right)$$

$$G(\sigma) = n_A(\sigma)\mu_A(\sigma) + n_B(\sigma)\mu_B(\sigma) + A\gamma$$

Step 2: Write down the total derivative of $G(\sigma)$

$$dG(\sigma) = n_A(\sigma)d\mu_A(\sigma) + \mu_A(\sigma)dn_A(\sigma) + n_B(\sigma)d\mu_B(\sigma) + \mu_B(\sigma)dn_B(\sigma) + \gamma dA + Ad\gamma$$

$$dG(\sigma) = -S(\sigma)dT + V(\sigma)dp + \mu_A(\sigma)dn_A(\sigma) + \mu_B(\sigma)dn_B(\sigma) + \gamma dA$$

$$dG(\sigma) = \mu_A(\sigma)dn_A(\sigma) + \mu_B(\sigma)dn_B(\sigma) + \gamma dA \qquad (T, P)$$

$$n_A(\sigma)d\mu_A(\sigma) + n_B(\sigma)d\mu_B(\sigma) + Ad\gamma = 0$$

This the Gibbs-Duhem equation of surface phase.

Step 3: Use Gibbs-Duhem equation of bulk phase

$$n_A(\mathrm{sln})d\mu_A(\mathrm{sln}) + n_B(\mathrm{sln})d\mu_B(\mathrm{sln}) = 0$$

$$d\mu_A(\mathrm{sln}) = -\frac{n_B(\mathrm{sln})}{n_A(\mathrm{sln})}d\mu_B(\mathrm{sln})$$

Step 4: At equilibrium, for A and B: $\mu(\mathrm{sln}) = \mu(\sigma)$

$$n_A(\sigma)\left(\frac{n_B(\mathrm{sln})}{n_A(\mathrm{sln})}d\mu_B\right) + n_B(\sigma)d\mu_B + Ad\gamma = 0$$

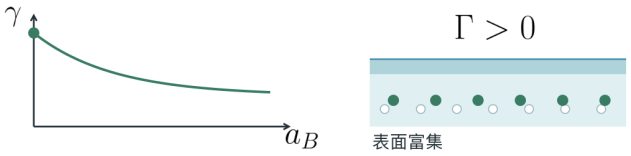
$$d\gamma = -\left[\frac{n_A(\sigma) - n_A(\sigma)\frac{n_B(\mathrm{sln})}{n_A(\mathrm{sln})}}{A}\right]d\mu_B \qquad d\gamma = -\Gamma d\mu_B$$

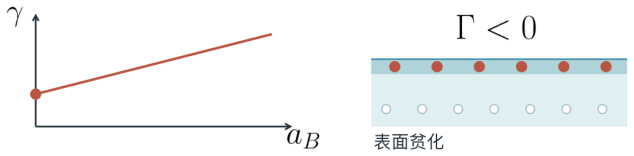
$$d\mu_B = d\left(\mu_B^{\ominus} + RT\ln a_B + \int_{p^{\ominus}}^{p} V_B^{\infty}dp\right) \qquad (T, P)$$

$$= d(RT\ln a_B) = \left(\frac{RT}{a_B}\right)\cdot da_B$$

$$\Gamma = -\frac{a_B}{RT}\left(\frac{\partial\gamma}{\partial a_B}\right)_{T,p}$$

Gibbs surface excess





Gibbs Surface Excess

$$\Gamma = -\frac{a_B}{RT}\left(\frac{\partial \gamma}{\partial a_B}\right)_{T,p}$$ Gibbs surface excess

If the activity coefficient is supposed to be 1, we can get another two expressions:

$$\Gamma = -\frac{b_B}{RT}\left(\frac{\partial \gamma}{\partial b_B}\right)_{T,p} \qquad \Gamma = -\frac{c_B}{RT}\left(\frac{\partial \gamma}{\partial c_B}\right)_{T,p}$$

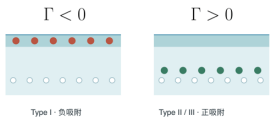
Take surface adsorption of water for example:

For Type I: $\frac{c_B}{RT}\left(\frac{\partial \gamma}{\partial c_B}\right)_{T,p} > 0$, thus $\Gamma < 0$.

We call this phenomenon as negative adsorption (负吸附)

For Type II and Type III: $\frac{c_B}{RT}\left(\frac{\partial \gamma}{\partial c_B}\right)_{T,p} < 0$, thus $\Gamma > 0$.

We call this phenomenon as negative adsorption (正吸附)



Depending on the type of solute, the change of surface tension of water behaves differently:

Type I: inorganic salts, which slightly increase the surface tension of water.

Type II: organics, which reduce the surface tension of water, with an approximate limit.

Type III: surfactants, which greatly reduce the surface tension of water even with a low concentration.

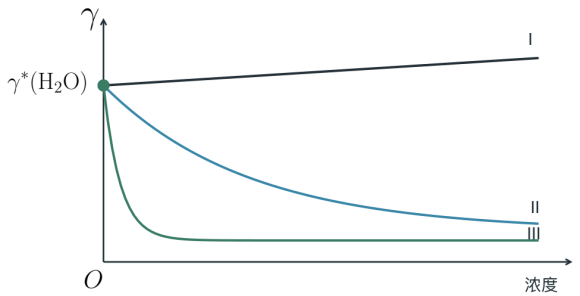


图 11-24　水溶液的表面张力与浓度的关系

Gibbs Surface Excess

For aqueous solution of some organic molecules, surface tension of water can be described by the following two-constant empirical equation:

$$\frac{\gamma^*(\mathrm{H_2O})-\gamma}{\gamma^*(\mathrm{H_2O})}=b\ln\left(\frac{c_B}{a}+1\right)$$ Syszkowski equation

The Gibbs surface excess thus can be derived:

$$-\left(\frac{\partial\gamma}{\partial c_B}\right)_{T,p}=\frac{b\gamma^*(\mathrm{H_2O})}{c_B+a}$$

$$-\left(\frac{\partial\gamma}{\partial\left(\frac{c_B}{c^\ominus}\right)}\right)_{T,p}=\frac{b\gamma^*(\mathrm{H_2O})}{\frac{c_B}{c^\ominus}+\frac{a}{c^\ominus}}$$

$$-\left(\frac{\partial\gamma}{\partial a_B}\right)_{T,p}=\frac{b\gamma^*(\mathrm{H_2O})}{a_B+\frac{a}{c^\ominus}}\qquad a'=\frac{a}{c^\ominus}$$

$$\Gamma=\frac{b\gamma^*(\mathrm{H_2O})}{RT}\cdot\frac{a_B}{a_B+a'}$$

At a given temperature, $\frac{b\gamma^*(\mathrm{H_2O})}{RT}$

is a constant, and assume

$$\Gamma_{\max}=\frac{b\gamma^*(\mathrm{H_2O})}{RT}\quad:\quad\Gamma=\frac{\Gamma_{\max}\cdot a_B}{a_B+a'}$$

- At low c_B ($a_B=\frac{c_B}{c^\ominus}$ is small), $a_B+a'\approx a'$:

$$\Gamma=\frac{\Gamma_{\max}\cdot a_B}{a'}$$

- At high c_B, $a_B+a'\approx a_B$: $\Gamma=\Gamma_{\max}$

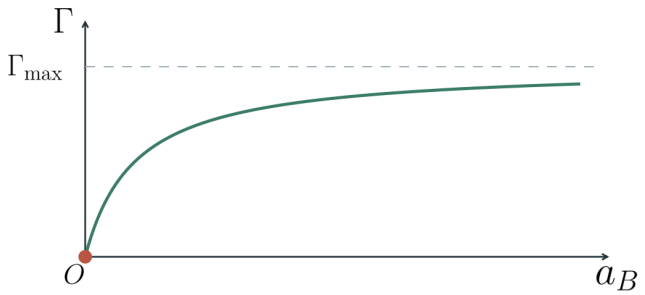


图 11-26 吸附等温线

An Example

例 11-2 $19°C$ 时丁酸水溶液的表面张力可以表达为 $\gamma = \gamma^*(H_2O) - a\ln(1+bc_B)$，其中 $\gamma^*(H_2O)$ 是纯水的表面张力，a 和 b 为常数。

(1) 试求此溶液中丁酸的表面吸附量 Γ 与浓度 c_B 的关系式；

(2) 若 $a = 0.0131\,\mathrm{N\cdot m^{-1}}$，$b = 0.0962\,\mathrm{m^3\cdot mol^{-1}}$ 试计算 $c_B = 200\,\mathrm{mol\cdot m^{-3}}$ 时的 Γ；

(3) 求饱和吸附量 $\Gamma_{\max}$，若此时表面层的丁酸分子成单分子层吸附，试计算丁酸分子（在液面上）的截面积。

$$\Gamma = \frac{abc_B}{RT(1+bc_B)}$$

当 c_B 很大时，$bc_B \gg 1$，即 $(1+bc_B) \approx bc_B$

此时 $\Gamma = a/RT$，表明 Γ 与 c_B 无关，即为饱和吸附，所以

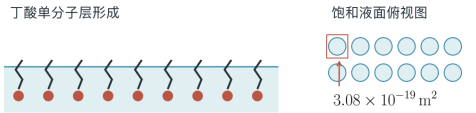
$$\Gamma_{\max} = \frac{a}{RT} = \frac{0.0131}{8.314\times 292.2}\,\mathrm{mol\cdot m^{-2}}$$

$$= 5.39\times 10^{-6}\,\mathrm{mol\cdot m^{-2}}$$

虽然 Γ 的原意是表面超量，但由于饱和吸附时，表面活性物质在表面层的浓度远较没有吸附时大，所以 $\Gamma_{\max}$ 也可当做单位表面上溶质的总物质的量。

此结果表明，当丁酸分子在液面上恰盖满一层时，$1\,\mathrm{m^2}$ 表面上将有 $5.39\times 10^{-6}\times 6.023\times 10^{23}$ 个丁酸分子，所以丁酸分子的截面积为

$$\frac{1}{5.39\times 10^{-6}\times 6.023\times 10^{23}}\mathrm{m^2} = 3.08\times 10^{-19}\,\mathrm{m^2}$$

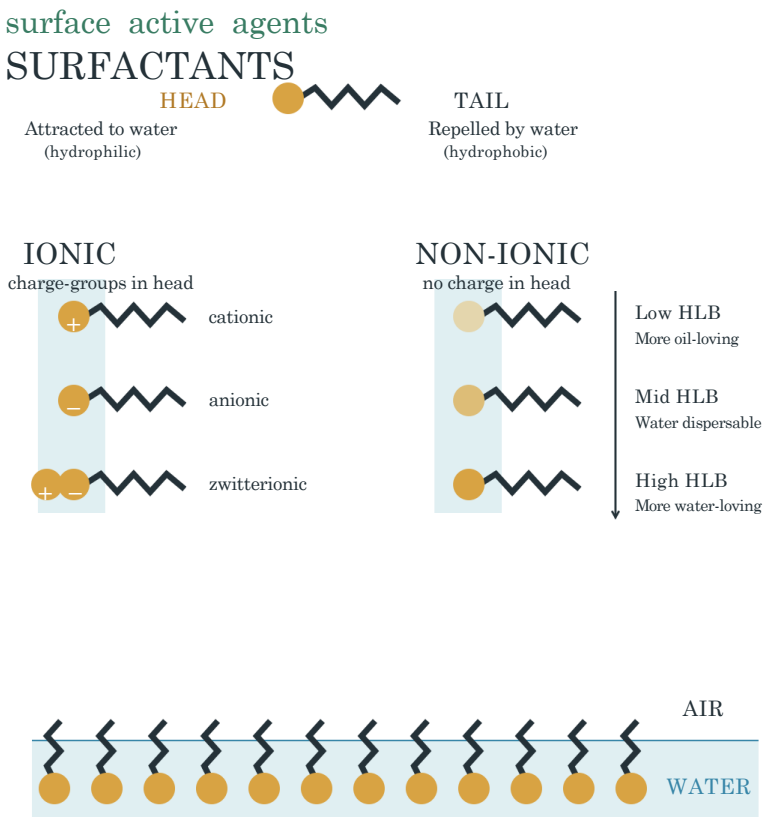


Surfactants

- Surfactants (表面活性剂) are type of molecule that effectively lowers the surface tension between two materials, either a gas and a liquid (e.g. water surface tension), two liquids (e.g. water and oil), or a liquid and solid (water and dirt particles).

- All surfactants are molecules that consist of a hydrophilic "water loving" (亲水性) head, and a hydrophobic "water hating" (疏水性) tail. In more technical terms, surfactants are amphiphilic molecules (两亲性分子), meaning they are molecules that contain both a hydrophilic and hydrophobic (疏水性) region.

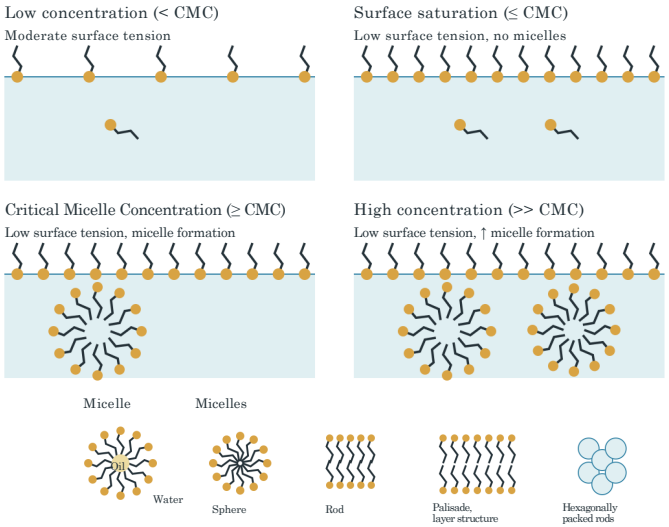
- Surfactants assemble (排列), typically with orientation, on the surface of the water to form a layer at the water-air interface, with the hydrophilic heads towards the water, and the hydrophobic tails in the air. This results in a decrease in surface tension – a key function of surfactants.



https://www.dispersa.ca/blog/what-are-surfactants-and-how-do-they-work/

Micelle

- When surfactant concentration increases, micelles (胶束) form in liquid phase, with inward pointing tails away from the water. The surfactant concentration at which micelles begin to form is called the critical micelle concentration (CMC) (临界胶束浓度). Adding more surfactant beyond the CMC results in increased micelle formation, while surface tension at the water-air interface remains low.



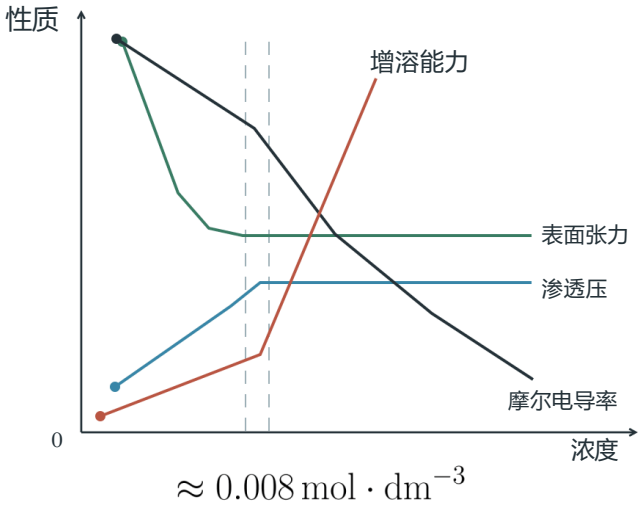
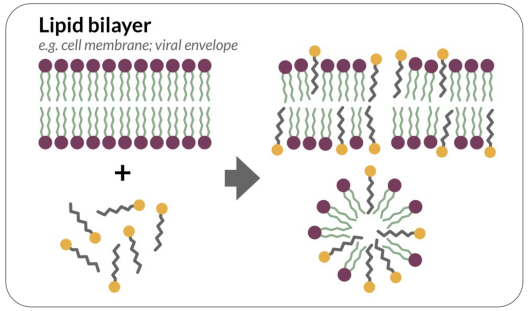


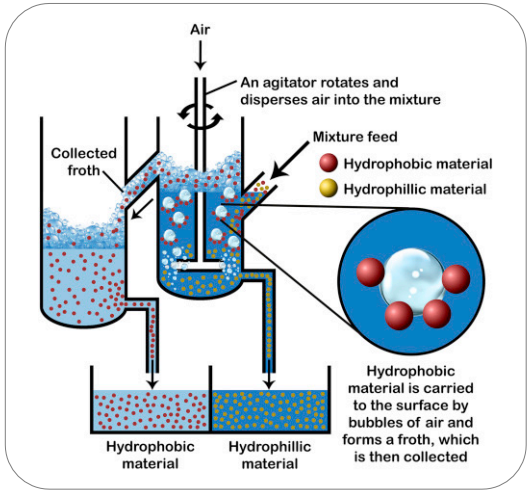
图 11-30　十二烷基磺酸钠水溶液的各种性质与浓度的关系

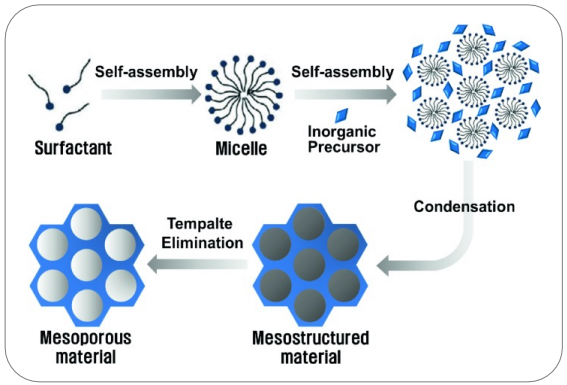
Applications of Surfactants

- Surfactants find wide applications in our daily life and industry. Below are some of the typical functions:
  - As detergents and disinfectants.
  - Optimizing the wetting of solid surfaces.
  - Connection between water and hydrophobic phases.
  - Improvement of dispersibility of powders/droplets.
  - Foam formation and foam prevention.
  - Templates for synthesizing nanomaterials.



Surfactants in hand soap can break down the lipid membranes surrounding bacteria and enveloped viruses like those causing COVID-19 and the flu, simultaneously destroying the microbes while facilitating their removal from your hands.

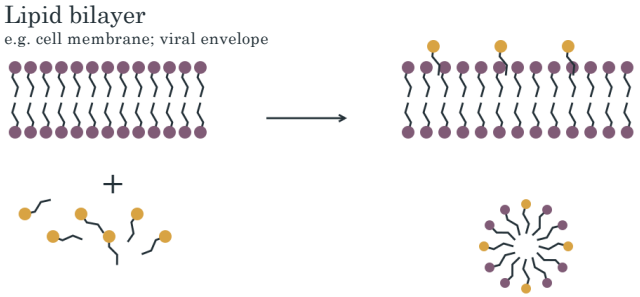




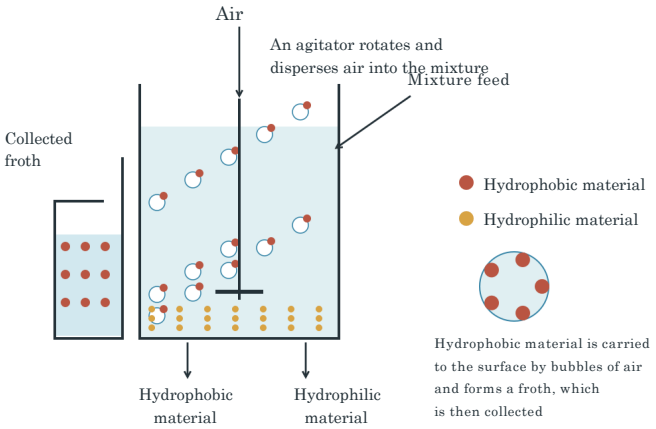
Applications of Surfactants

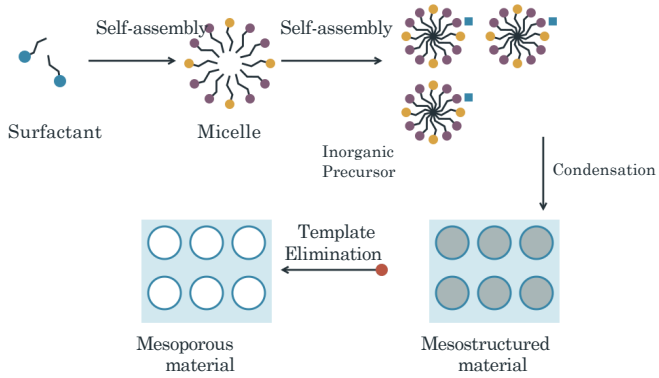
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Solid-Liquid Interface: Wetting

- Due to higher inhomogeneity on solid surface, the solids typically have a higher surface tension than the liquids.

  Surface tension of a liquid: $< 0.1\,\mathrm{N \cdot m^{-1}}$

  Solid surface tension (low energy surface): $< 0.1\,\mathrm{N \cdot m^{-1}}$

  Solid surface tension (high energy surface): $> 0.1\,\mathrm{N \cdot m^{-1}}$

- As the solid surface has a higher surface energy, it tends to attract other molecules, either liquids or gases.

- When a solid-gas interface and a liquid-gas interface meet, it is spontaneous that the two interfaces are replaced by a solid-liquid interface. This process is called wetting (润湿).

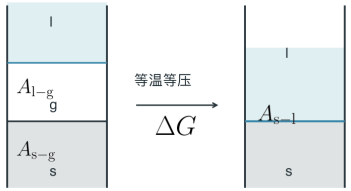


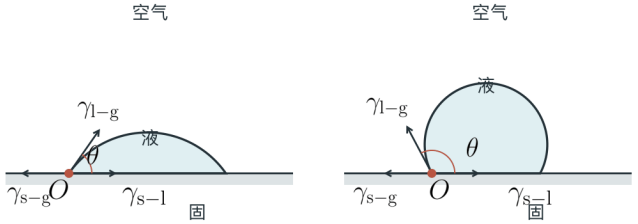
图 11-15　润湿过程

$$\Delta G = \gamma_{s-l}A_{s-l} - (\gamma_{l-g}A_{l-g} + \gamma_{s-g}A_{s-g})$$

$$\frac{\Delta G}{m^2} = \gamma_{s-l} - (\gamma_{l-g} + \gamma_{s-g})$$

ΔG can be obtained from above equation. The problem is that γ_{s-l} and γ_{s-g} are different to determine.

- Contact angle (接触角) is geometrically defined as the angle formed by a liquid at the three-phase boundary.



$$\gamma_{s-g} = \gamma_{s-l} + \gamma_{l-g}\cos\theta$$ Young equation

$$\frac{\Delta G}{m^2} = -\gamma_{l-g}(1+\cos\theta)$$

$$\frac{d\frac{\Delta G}{m^2}}{d\theta} = \gamma_{l-g}\sin\theta > 0$$

$\theta > 90°$ non-wetting (不润湿)

$\theta < 90°$ wetting (润湿)

$\theta = 90°$ complete wetting (完全润湿)

Solid-Liquid Interface: Spreading

- If a liquid is dropped on a solid surface and spontaneously spreads into a film, this process is called spreading (铺展).

Surface tension of a liquid: $< 0.1\,\mathrm{N \cdot m^{-1}}$

Solid surface tension (low energy surface): $< 0.1\,\mathrm{N \cdot m^{-1}}$

Solid surface tension (high energy surface): $> 0.1\,\mathrm{N \cdot m^{-1}}$

For metals and glasses (high energy surface): $0.1\text{–}1.0\,\mathrm{N \cdot m^{-1}}$

- It is obvious that water tends to spread on surfaces such as metal and glass, while it does not on organic surfaces such as paraffin and PTFE whose surface tension is $< 0.1\,\mathrm{N \cdot m^{-1}}$.

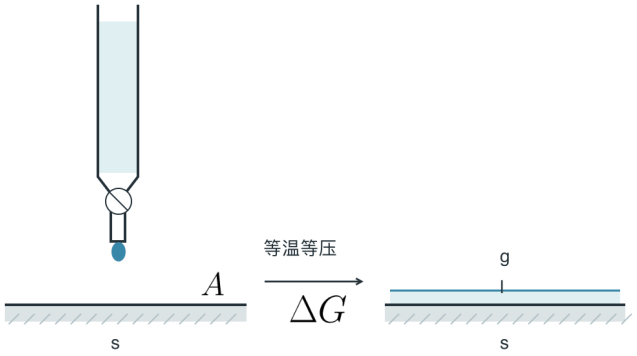


图 11-17　铺展过程

$$\Delta G = \gamma_{\mathrm{s-l}} A_{\mathrm{s-l}} + \gamma_{\mathrm{l-g}} A_{\mathrm{l-g}} - \gamma_{\mathrm{s-g}} A_{\mathrm{s-g}}$$

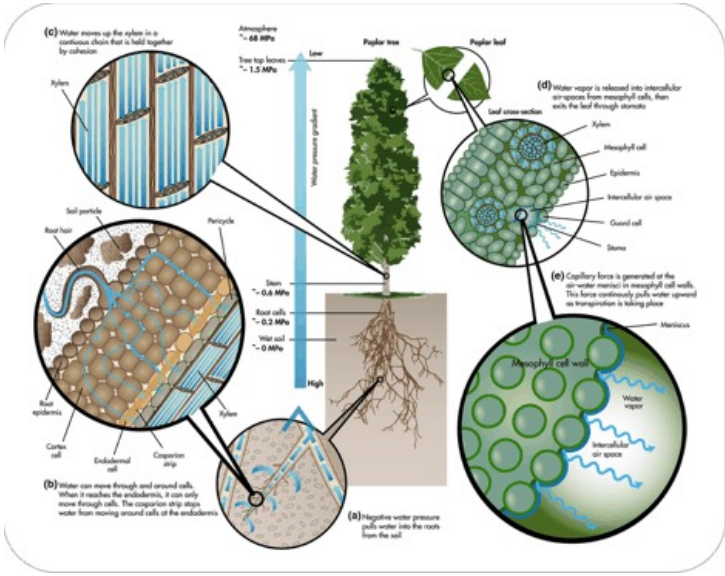
$$\frac{\Delta G}{m^2} = \gamma_{\mathrm{s-l}} + \gamma_{\mathrm{l-g}} - \gamma_{\mathrm{s-g}}$$

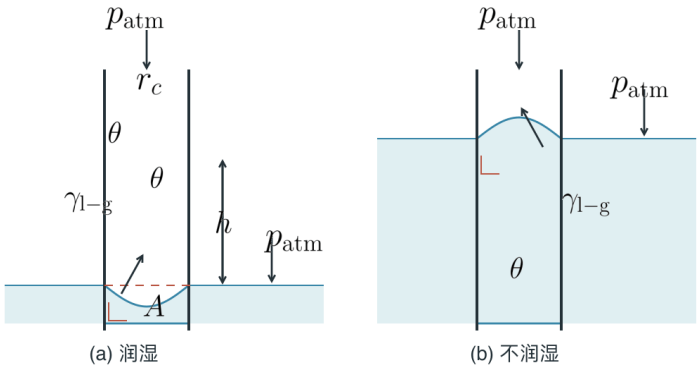
If the Young equation applies, we have:

$$\frac{\Delta G}{m^2} = \gamma_{\mathrm{l-g}}(1 - \cos\theta)$$

The Capillary Action

- If a capillary tube is placed vertically into a liquid, capillary action (or capillary effect) (毛细作用/毛细现象) will raise or suppress the liquid inside the tube depending on the combination of substances. The actual effect depends on the relative strength of the cohesive and adhesive forces and, thus, the contact angle.





$$p_{\text{upward}} = p_{atm} + \Delta p$$

$$p_{\text{downward}} = p_{atm} + \rho g h$$

$$\Delta p = \frac{2\gamma_{l-g}}{r} = \rho g h \qquad r = \frac{r_c}{\cos\theta}$$

$$\gamma_{l-g} = \frac{\rho g h r_c}{2\cos\theta}$$

This method can be used to determine γ_{l-g}.

Solid Surface

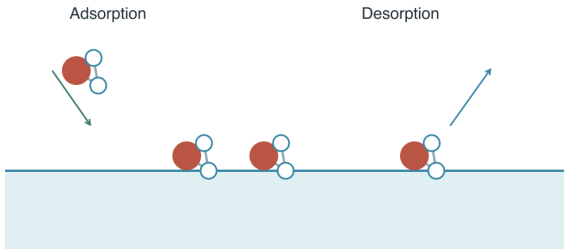
- Solid surfaces have the following features:

  The surface molecules are not mobile.

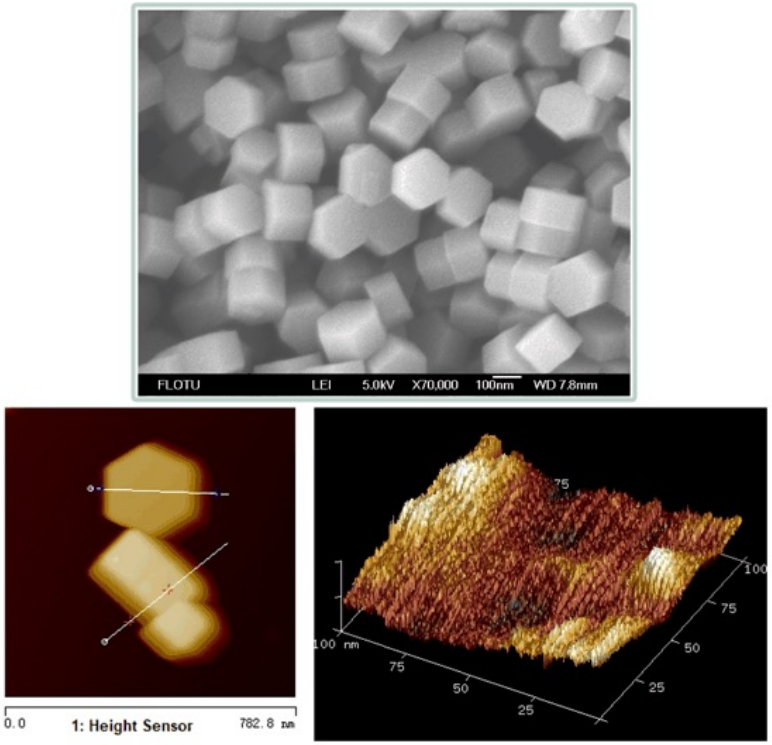
  The solids typically have a higher surface tension than the liquids.

  The solid surface is not flat but typically has a higher inhomogeneity.

- Solids have a higher surface energy and thus they can adsorb gas molecules spontaneously to minimize the surface energy. This process is adsorption (吸附). The adsorbed molecules can go back to the gas phase due to thermal motion, which is called desorption (脱附).

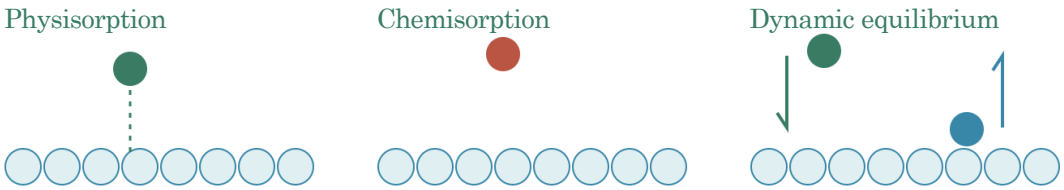


zeolite crystals and surfaces



Adsorption

- Adsorption can be physical (physisorption) (物理吸附) or chemical (chemisorption) (化学吸附). The adsorbing solid is known as adsorbent (吸附剂), and the material in the adsorbed state is known as adsorbate (吸附质).
- Physisorption occurs whenever an adsorbable gas is brought into contact with the surface of a solid. The intermolecular forces involved are of the same kind as those responsible for the imperfection of real gases and the condensation of vapors.
- Chemisorption refers to the adsorption due to the intermolecular forces which lead to the formation of chemical bonds.
- When the adsorption rate equals to the desorption rate, an equilibrium is reached. The adsorption capacity at equilibrium can be described by volumetric adsorption amount ($\Gamma/\mathrm{mol}\cdot\mathrm{kg}^{-1}$) or fractional coverage of the solid surface (θ). the expression describing its variation with pressure at a chosen temperature is called the adsorption isotherm (吸附等温线).



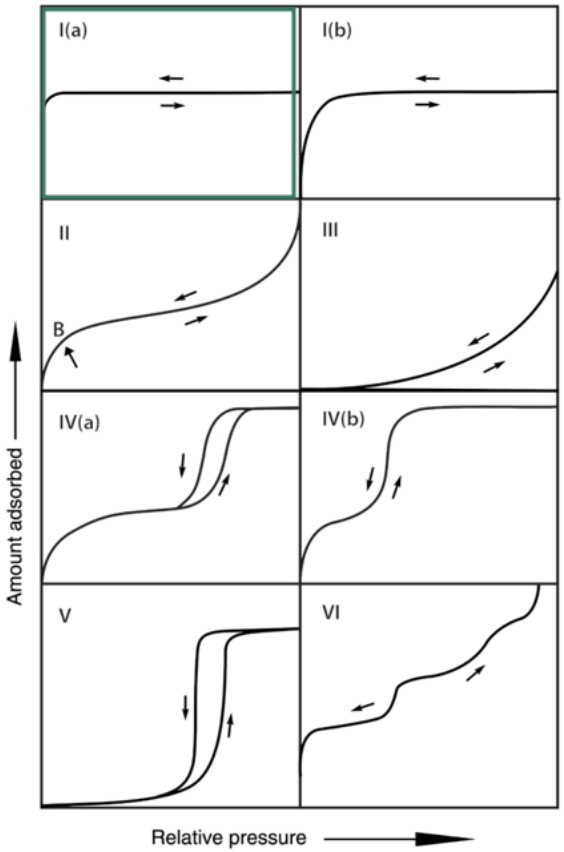


Fig. 2: Classification of physisorption isotherms.

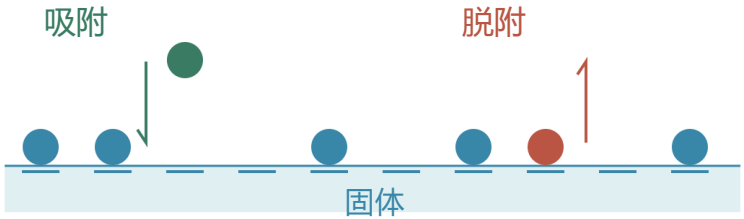
The Langmuir Isotherm

- An adsorption theory proposed by Irving Langmuir is the simplest that is physically plausible, which is based on the following assumptions:

  Adsorption cannot exceed monolayer coverage.

  All sites on the surface are equivalent. A molecule can be adsorbed only at a vacant site.

  The probability of adsorption is independent of the occupation of neighboring sites (that is, there is no interaction among adsorbed molecules).



θ : 表面覆盖率　　$1-\theta$: 表面空白率

$$r_{\mathrm{de}} \propto \theta \quad \Longrightarrow \quad r_{\mathrm{de}} = k_{\mathrm{de}}\theta$$

$$r_{\mathrm{ad}} \propto p \qquad r_{\mathrm{ad}} \propto (1-\theta)$$

$$r_{\mathrm{ad}} = k_{\mathrm{ad}} p\,(1-\theta)$$

$$B\,(\mathrm{g}) + A \overset{\text{equilibrium}}{\rightleftharpoons} AB$$

At equilibrium:

$$r_{\mathrm{ad}} = r_{\mathrm{de}}$$

$$k_{\mathrm{de}}\theta = k_{\mathrm{ad}} p\,(1-\theta)$$

$$\theta = \frac{k_{\mathrm{ad}}p}{k_{\mathrm{de}}+k_{\mathrm{ad}}p} = \frac{\frac{k_{\mathrm{ad}}}{k_{\mathrm{de}}}p}{1+\frac{k_{\mathrm{ad}}}{k_{\mathrm{de}}}p}$$

$$b = \frac{k_{\mathrm{ad}}}{k_{\mathrm{de}}}$$

$$\theta = \frac{bp}{1+bp}$$

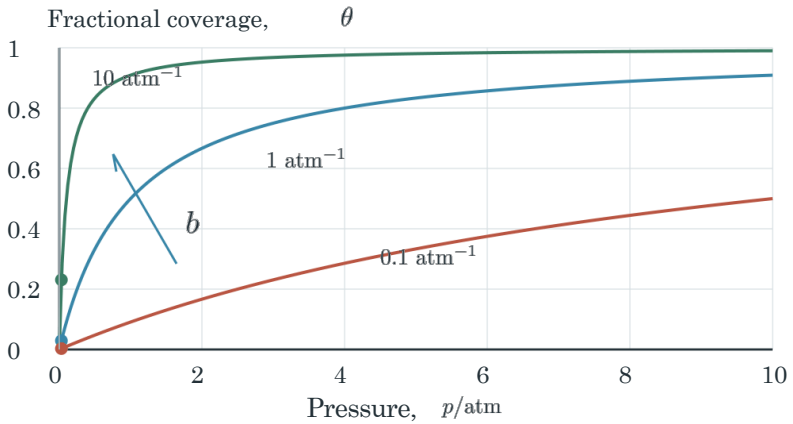
Langmuir isotherm

The Langmuir Isotherm

Use Γ_{max} to represent the adsorption amount at $\theta = 1$. At any adsorption equilibrium, we have:

$$\Gamma = \Gamma_{max}\theta \quad \rightarrow \quad \Gamma = \frac{\Gamma_{max} \cdot bp}{1+bp}$$

- At low pressure, $bp \ll 1$: $\Gamma = \Gamma_{max} bp$
- At high pressure, $bp \gg 1$: $\Gamma = \Gamma_{max}$



If we use V_{max} to represent the adsorption amount at $\theta = 1$, then we have:

$$V = \frac{V_{max} \cdot bp}{1+bp} \quad \rightarrow \quad V + bpV = \Gamma_{max} \cdot bp$$

$$\frac{p}{V} = \frac{p}{V_{max}} + \frac{1}{bV_{max}}$$

- The above equation can be used to determine the surface area of an adsorbent, because V_{max} represents the amount of adsorbates that cover the solid surface in a monolayer manner.
- According to its definition ($b = \frac{k_{ad}}{k_{de}}$), b is a function of T but not relevant with p.
- It is possible to relate the temperature dependence of b to the isosteric enthalpy of adsorption.

The Langmuir Isotherm

The quantity b is the ratio of the rate constants for the forward and reverse reactions in the equilibrium $B(\mathrm{g}) + A(\mathrm{surface}) \rightleftharpoons AB(\mathrm{surface})$, thus:

Step 1: Relate b to the equilibrium constant ($K^{\ominus}$)

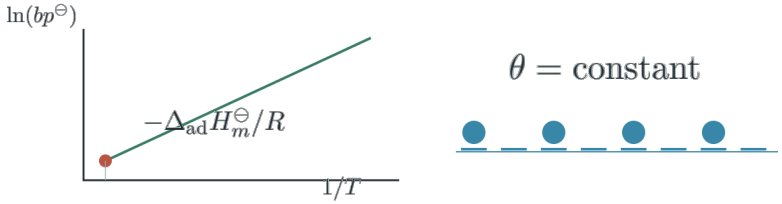
$$K^{\ominus} = \frac{\theta}{(1-\theta)\cdot\left(\frac{p}{p^{\ominus}}\right)} \qquad b = \frac{k_{\mathrm{ad}}}{k_{\mathrm{de}}} = \frac{\theta}{p(1-\theta)}$$

$$K^{\ominus} = bp^{\ominus}$$

Step 2: Relate b to the standard Gibbs energy of adsorption ($\Delta_{\mathrm{ad}}G^{\ominus}$)

$$\Delta_r G_m^{\ominus} = -RT\ln K^{\ominus} \quad \rightarrow \quad \Delta_{\mathrm{ad}} G_m^{\ominus} = -RT\ln K^{\ominus}$$

$$-R\ln(bp^{\ominus}) = \frac{\Delta_{\mathrm{ad}}G_m^{\ominus}}{T}$$



Step 3: Use the Gibbs-Helmholtz equation to the temperature dependence of $\frac{\Delta_{\mathrm{ad}}G_m^{\ominus}}{T}$ to the enthalpy of adsorption ($\Delta_{\mathrm{ad}}G_m^{\ominus}$):

$$\left[\frac{\partial\left(\frac{\Delta G}{T}\right)}{\partial T}\right]_p = -\frac{\Delta H}{T^2} \qquad \frac{d\ln K^{\ominus}}{dT} = \frac{\Delta_r H_m^{\ominus}}{RT^2}$$

$$\Delta_r G_m^{\ominus} = -RT\ln K^{\ominus}$$

$$\frac{d\ln(bp^{\ominus})}{dT} = \frac{\Delta_{\mathrm{ad}}H_m^{\ominus}}{RT^2} \qquad \left[\frac{\partial\ln(bp^{\ominus})}{\partial T}\right]_{\theta} = \frac{\Delta_{\mathrm{ad}}H_m^{\ominus}}{RT^2}$$

$$\left[\frac{\partial\ln(bp^{\ominus})}{\partial(1/T)}\right]_{\theta} = -\frac{\Delta_{\mathrm{ad}}H_m^{\ominus}}{R}$$

isosteric adsorption enthalpy

- The standard enthalpy of adsorption depends on the fractional coverage, so this expression is restricted to constant θ . $\Delta_{\mathrm{ad}}H_m^{\ominus}$ must be interpreted as the isosteric enthalpy of adsorption.

The BET Theory

- If the initial adsorbed layer can act as a substrate for further adsorption, then instead of the volume of gas adsorbed levelling off at high pressures to a value corresponding to a complete monolayer, it can be expected to rise indefinitely. The most widely used isotherm dealing with multilayer adsorption was derived by Stephen Brunauer, Paul Emmett, and Edward Teller and is called the BET isotherm:

  Multiple-layered adsorption is possible: The multiple layers can start simultaneously rather than one by one. Van der Waals interaction is responsible for the second layer adsorption and above.

  All sites on the surface are equivalent. A molecule can be adsorbed only at a vacant site.

  The probability of adsorption is independent of the occupation of neighboring sites (that is, there is no interaction among adsorbed molecules).

$$V = \frac{CpV_{\max}}{(p_v - p)\left[1 + \frac{(C-1)p}{p_v}\right]}$$

BET equation

$V_{\max}$: the volume of gas corresponding to a complete monolayer

C : a constant relevant to adsorption enthalpy

$$\frac{p}{V(p_v - p)} = \frac{C-1}{V_{\max}C} \cdot \frac{p}{p_v} + \frac{1}{V_{\max}C}$$



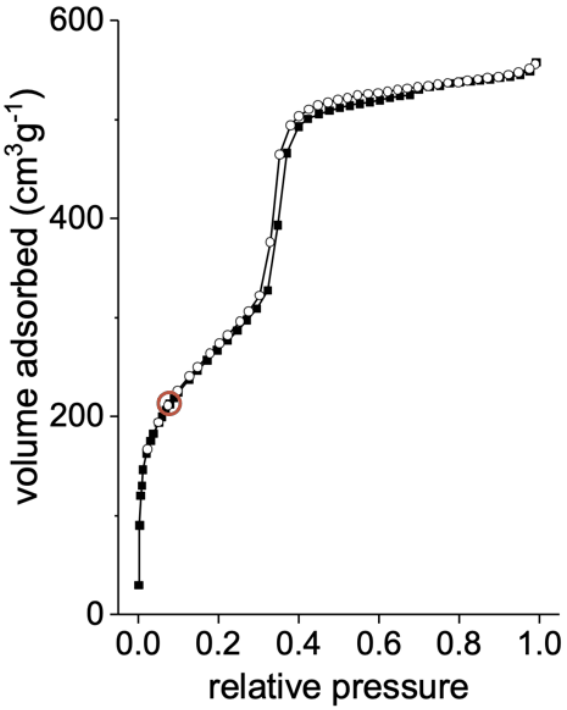
- The equation is widely used to determine surface area of an adsorbent.

Gas Adsorption Measurement



BET Surface Area

例题：在77.2K时以 N_2 为吸附质，测得某二氧化硅材料的氮气吸附-脱附等温线如下图所示（吸附数据见表格）。77.2K时液氮的饱和蒸气压为99125Pa， N_2 分子的截面积为 $16.2 \times 10^{-20}\,\mathrm{m}^2$ ，试用BET公式计算该材料的比表面积。



p/p_v	V (cm^3/g, STP)	p/p_v	V (cm^3/g, STP)
0.012	146.09	0.450	505.41
0.021	163.02	0.477	508.63
0.030	175.13	0.502	511.34
0.038	182.57	0.527	513.94
0.051	193.93	0.553	515.79
0.059	199.75	0.578	517.25
0.068	206.22	0.603	519.44
0.079	212.50	0.628	521.56
0.089	218.41	0.653	523.38
0.099	223.87	0.678	524.70
0.127	237.54	0.700	530.10
0.147	246.51	0.727	532.76
0.171	256.65	0.753	533.81
0.196	266.76	0.778	536.13
0.222	276.92	0.803	537.71
0.247	286.89	0.829	538.22
0.271	297.17	0.853	539.68
0.296	308.85	0.878	540.56
0.323	327.43	0.903	541.71
0.349	393.15	0.928	543.28
0.371	465.93	0.953	545.29
0.401	492.76	0.976	548.94
0.424	500.23	0.993	557.66

BET Surface Area

解：

p/p_v	V (cm^3/g, STP)
0.079	212.50
0.147	246.51
0.222	276.93
0.296	308.86
0.323	327.43

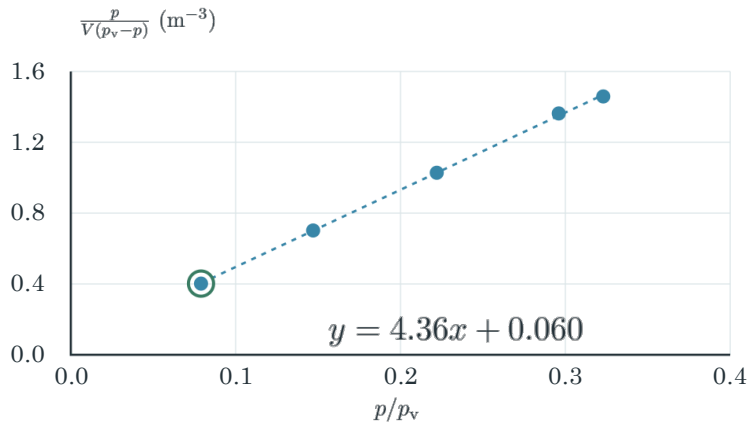
单位换算 ↓

p/p_v	V (m^3/g, STP)
0.079	0.213
0.147	0.247
0.222	0.277
0.296	0.309
0.323	0.327

求解 $p/[V(p_v-p)]$ ↓

p/p_v	$\frac{p}{V(p_v-p)}$ (m^{-3})
0.079	0.402
0.147	0.702
0.222	1.028
0.296	1.363
0.323	1.459

利用 $p/[V(p_v-p)]$ 对作图 p/p_v 作图，得一直线：



$(C-1)/V_{max}C$： 4.36 m^{-3}

$1/V_{max}C$： 0.060 m^{-3}

V_{max}： 0.226 m^3

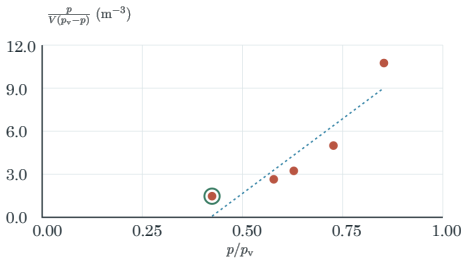
$$A_m = \frac{\left(\frac{V_{max}}{m}\right)\times L}{22.4\times10^{-3}m^3\cdot mol^{-1}} \times A_c(N_2)$$

$$= \frac{\left(\frac{0.226}{1}\right)\times 6.023\times10^{23}}{22.4\times10^{-3}} \times 16.2\times10^{-20}m^2\cdot kg^{-1}$$

$$= 980000 m^2\cdot kg^{-1}$$

relative pressure	volume (cm^3/g, STP)
0.424	500.2366
0.578	517.254
0.628	521.5698
0.727	532.7686
0.853	539.6893

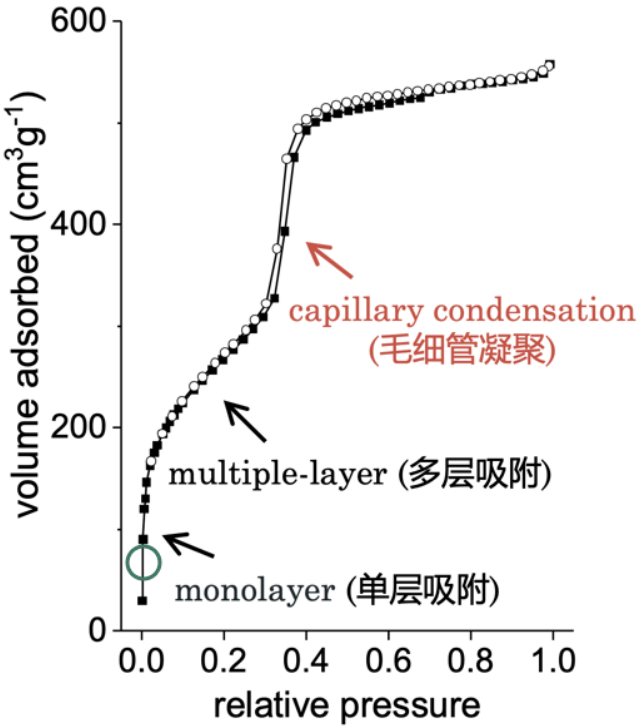
利用 $p/[V(p_v-p)]$ 对作图 p/p_v 作图，

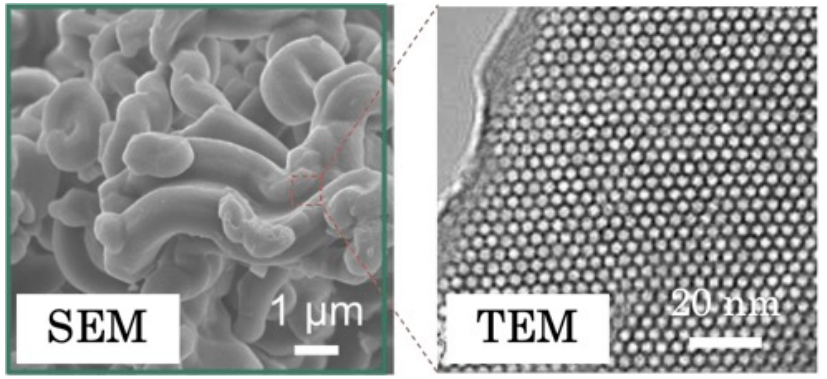


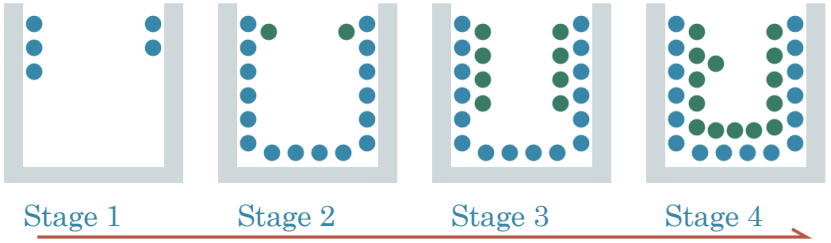
得不到一条直线……

N_2 Adsorption-Desorption Isotherms

BET 比表面积： $980\ m^2/g$

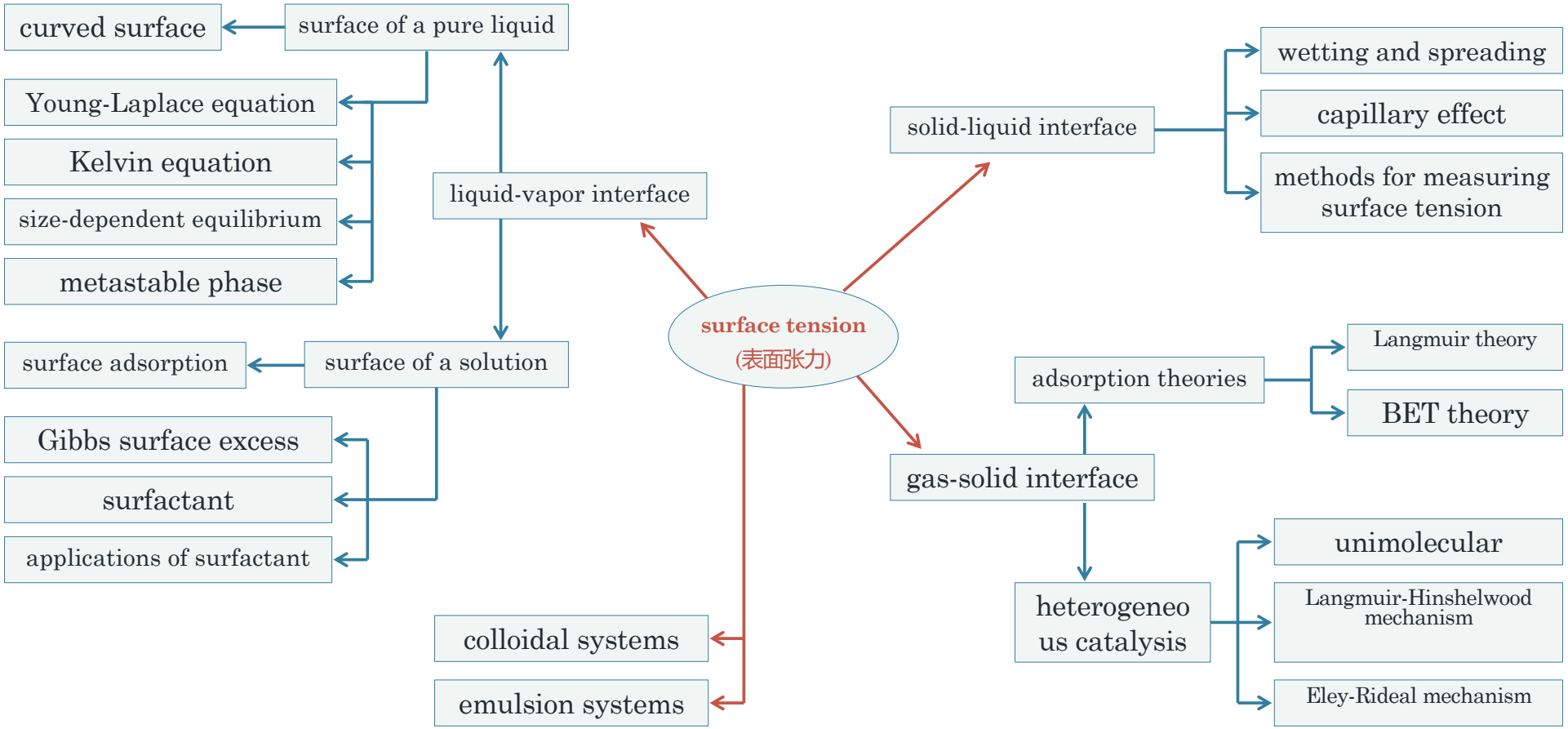






思考：为什么会发生毛细管凝聚？它和介孔孔径有什么关系？

A Mind Map for Surface Science



Physisorption versus Chemisorption



Physisorbed molecules　　Chemisorbed molecules

表 11-4　物理吸附与化学吸附的比较

性质	物理吸附	化学吸附
吸附力	Van der Waals 引力	化学键力
吸附热	近于液化热	近于化学反应热
选择性	无	有
吸附层数	单层或多层	单层
稳定性	不稳定，易脱附	稳定，不易脱附
吸附速率	较快，不受温度影响	较慢，温度升高速率加快
吸附温度 T	$T \ll$ 吸附质沸点	$T \gg$ 吸附质沸点
吸满单层的压力	$p/p_v \approx 0.1$	$p/p_v \ll 0.1$

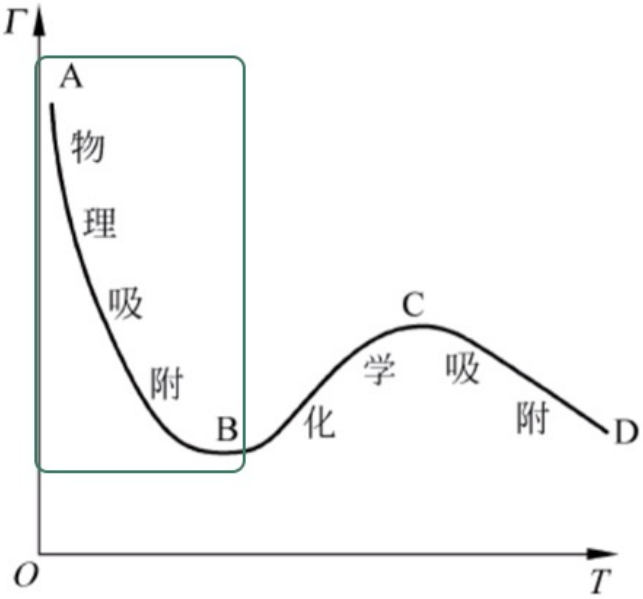


图 11-42　H_2 在 Ni 上的吸附等压线

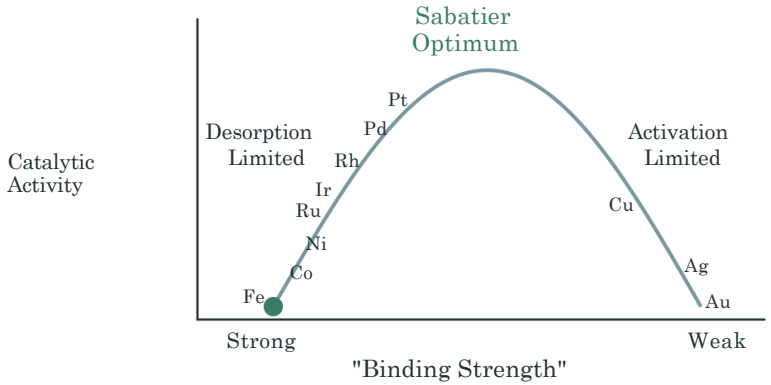
Physisorption is the adsorption in which the molecular interactions between the adsorbate molecules and the adsorbent are primarily governed by van der Waals forces, whereas chemisorption involves the formation of chemical bonds.

Chemisorption and Heterogeneous Catalysis

The heat of adsorption (吸附热) Q or $\Delta_{\mathrm{ad}}H_m$ is an indicator of the strength of the interaction between an adsorbate and a solid adsorbent.
The heat of adsorption can be determined from the heat released in calorimetric experiments or from the analysis of adsorption isotherms at different temperatures. The latter, called isosteric heats of adsorption (等量吸附热), are commonly used in the characterization of adsorbents.
From Clausius-Clapeyron equation we get:

$$\left(\frac{\partial \ln p}{\partial T}\right)_{\Gamma} = -\frac{\Delta_{\mathrm{ad}}H_m}{RT^2}$$

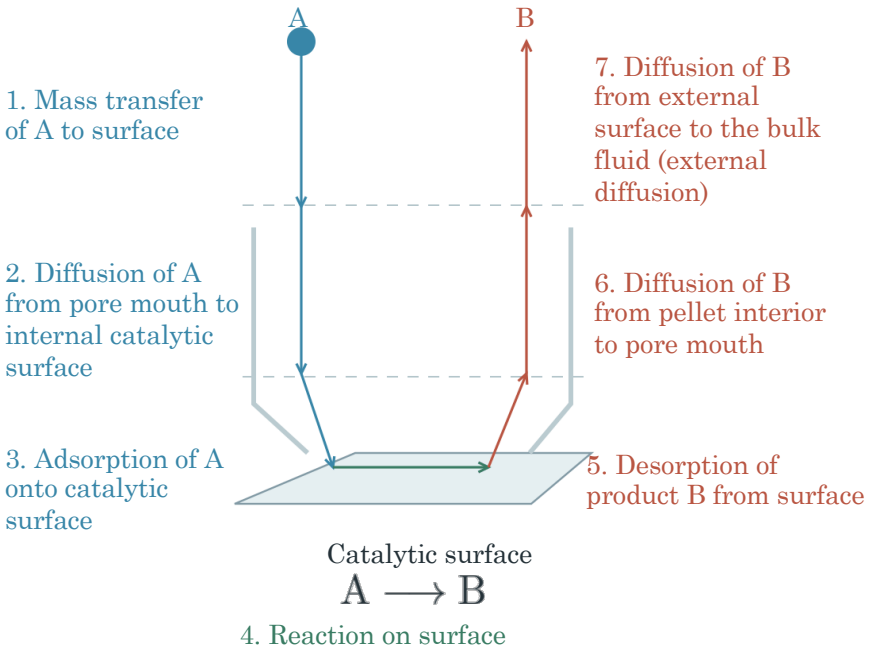
With the increase of Γ, $\Delta_{\mathrm{ad}}H_m$ generally decreases because of the heterogeneous surface sites.
The relative magnitudes of $\Delta_{\mathrm{ad}}H_m$ can be used to understand the process of adsorption.



The Sabatier principle is a qualitative concept in chemical heterogeneous catalysis, which states that the interactions between the catalyst and the substrate should be "just right"; that is, neither too strong nor too weak. If the interaction is too weak, the molecule will fail to bind to the catalyst and no reaction will take place. On the other hand, if the interaction is too strong, the product fails to dissociate. The principle can be shown graphically by plotting the reaction rate against a property such as the heat of adsorption of the reactant by the catalyst. Such plots pass through a maximum, looking roughly like a triangle or an inverted parabola, and are called volcano plots because of their shape.

Steps of Heterogeneous Catalysis

steps of heterogeneous catalysis



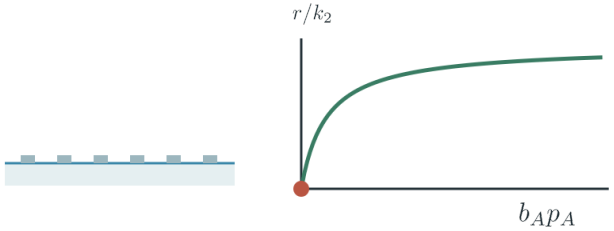
Heterogeneous catalysis has five stages:

Stage 1: Diffusion of Reactant(s) to the Surface: The rate at which reactants diffuse to the surface will be affected by their bulk concentration and by the thickness of the boundary layer.
Stage 2: Adsorption of reactants: Bonds are formed as the reactant(s) are adsorbed onto the surface of the catalyst. The ability for an atom or molecule to stick to the surface is known, brilliantly, as the sticking coefficient. This is just the ratio or percentage of molecules that end up sticking on the surface.
Stage 3: Surface reaction: Bonds form between the atoms and molecules on the surface
Stage 4: Desorption of products: Bonds are broken as the product(s) desorb from the surface.
Stage 5: Diffusion of Product(s) away from the Surface: The products are then desorbed from the surface of the catalyst.

Mechanism of Unimolecular Reactions

$$\mathrm{A} + \underset{-\mathrm{S}-}{|} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \underset{-\mathrm{S}-}{\overset{\mathrm{A}}{|}} \xrightarrow{k_2} \underset{-\mathrm{S}-}{\overset{\mathrm{B}}{|}} \underset{k_{-3}}{\overset{k_3}{\rightleftharpoons}} \underset{-\mathrm{S}-}{|} + \mathrm{B}$$

slow



If both reactant adsorption and product desorption are fast, the surface reaction is the rate-determining step (RDS), then the reaction rate is proportional to fractional coverage:

$$r = k_2\theta$$

Let's assume the Langmuir isotherm applies:

$$r = \frac{k_2 b_A p_A}{1+b_A p_A}$$

If the adsorption of B is not negligible:

$$r = \frac{k_2 b_A p_A}{1+b_A p_A+b_B p_B}$$

At low pressure, $b_A p_A \ll 1$, $r = k_2 b_A p_A$

The reaction behaves as a first-order reaction.

At high pressure, $b_A p_A \gg 1$, $r = k_2$

The reaction behaves as a zeroth-order reaction.

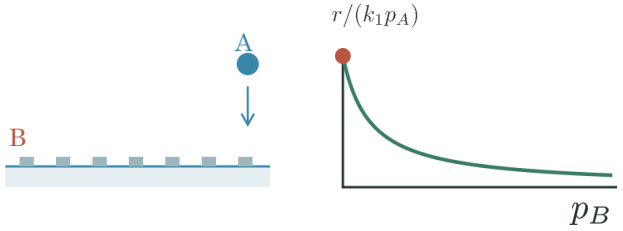
At medium pressure, $r = \frac{k_2 b_A p_A}{1+b_A p_A}$

The reaction order is not an integer.

This mechanism answers why a unimolecular reaction can behave as a zeroth-order reaction.

Mechanism of Unimolecular Reactions

$$\mathrm{A} + \underset{-\mathrm{S}-}{|} \xrightarrow[\textit{slow}]{k_1} \underset{-\mathrm{S}-}{\overset{\mathrm{A}}{|}} \underset{k_{-2}}{\overset{k_2}{\rightleftharpoons}} \underset{-\mathrm{S}-}{\overset{\mathrm{B}}{|}} \underset{k_{-3}}{\overset{k_3}{\rightleftharpoons}} \underset{-\mathrm{S}-}{|} + \mathrm{B}$$



If the reactant adsorption is the rate-determining step (RDS), the reaction rate is proportional to the pressure of A and the fraction of vacant sites:

$$r = k_1 p_A (1 - \theta_A - \theta_B)$$

Because the reactant adsorption is slow, the adsorption equilibrium is not reached for A, we should use the equilibrium pressure (p'_A) in the Langmuir equation, not the actual pressure (p_A). While for B, equilibrium is assumed.

$$\theta_A = \frac{b_A p'_A}{1 + b_A p'_A + b_B p_B} \qquad \theta_B = \frac{b_B p_B}{1 + b_A p'_A + b_B p_B}$$

The surface reaction is assumed at equilibrium:

$$k_2 \theta_A = k_{-2} \theta_B \quad \rightarrow \quad p'_A = \frac{b_B k_{-2}}{b_A k_2} p_B$$

$$r = \frac{k_1 p_A}{1 + \left(\frac{b_B k_{-2}}{b_A k_2} + 1\right) b_B p_B}$$

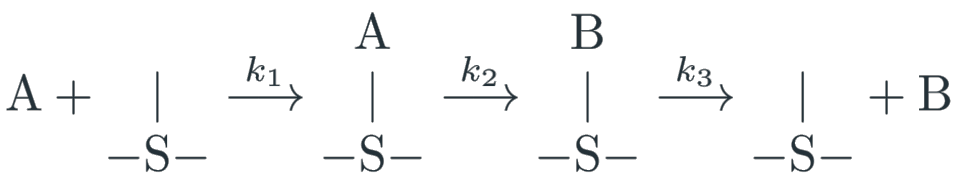
If b_B is small, $1 + \left(\frac{b_B k_{-2}}{b_A k_2} + 1\right) b_B p_B \approx 1, \quad r = k_1 p_A$

If b_B is large, $1 + \left(\frac{b_B k_{-2}}{b_A k_2} + 1\right) b_B p_B \approx \left(\frac{b_B k_{-2}}{b_A k_2} + 1\right) b_B p_B$

$$r = \frac{k_1 p_A}{\left(\frac{k_{-2}}{k_2} + 1\right) b_B p_B} = k \frac{p_A}{p_B}$$

Similar approach can be used to write the reaction mechanism if the product desorption is the RDS.

Mechanism of Unimolecular Reactions





If there is no rate-determining step, no pre-equilibrium can be assumed, and thus the Langmuir equation cannot be applied to represent θ_A and θ_B .

We apply the steady-state approximation assuming that θ_A and θ_B are not changing with time when a steady state is reached:

$$\frac{d\theta_A}{dt} = \frac{d\theta_B}{dt} = 0$$

In this case, we can consider that the overall rate equals to the rate in each step, and thus:

$r = k_2\theta_A$

Write down the expressions of $\frac{d\theta_A}{dt}$ and $\frac{d\theta_B}{dt}$:

$$\frac{d\theta_A}{dt} = k_1 p_A(1 - \theta_A - \theta_B) - k_2\theta_A = 0$$

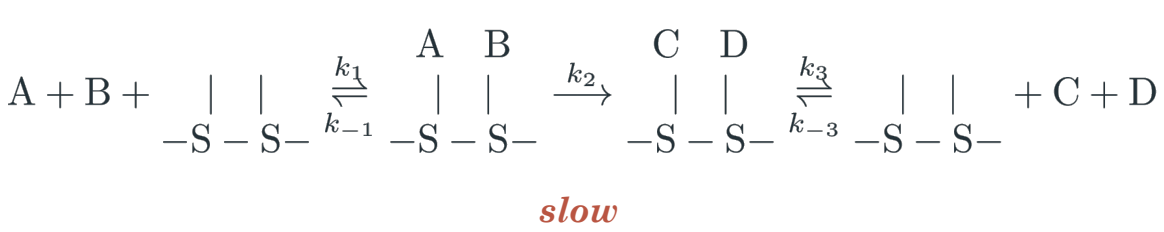
$$\frac{d\theta_B}{dt} = k_2\theta_A - k_3\theta_B = 0$$

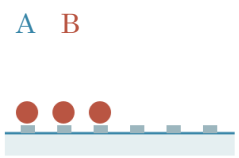
$$\theta_A = \frac{k_1 k_3 p_A}{k_2 k_3 + k_1(k_2 + k_3)p_A}$$

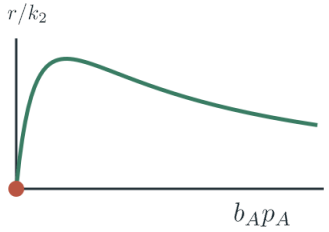
Rewrite the overall reaction rate:

$$r = k_2\theta_A = \frac{k_1 k_2 k_3 p_A}{k_2 k_3 + k_1(k_2 + k_3)p_A}$$

The Langmuir-Hinshelwood Mechanism







If both reactants adsorption and products desorption are fast, the surface reaction is the rate-determining step (RDS), then the reaction rate is proportional to fractional coverage:

$$r = k_2\theta_A\theta_B$$

Let's assume the Langmuir isotherm applies:

$$\theta_A = \frac{b_A p_A}{1+b_A p_A+b_B p_B}$$

$$\theta_B = \frac{b_B p_B}{1+b_A p_A+b_B p_B}$$

Rewrite the reaction rate:

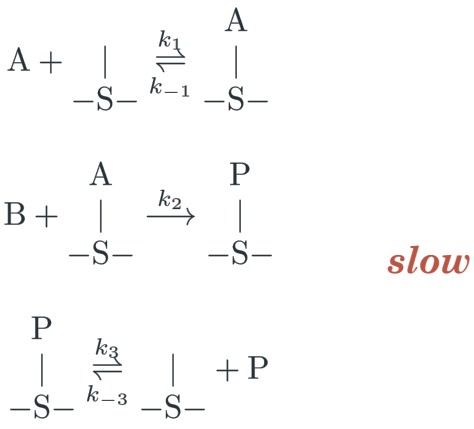
$$r = \frac{k_2 b_A b_B p_A p_B}{(1+b_A p_A+b_B p_B)^2}$$

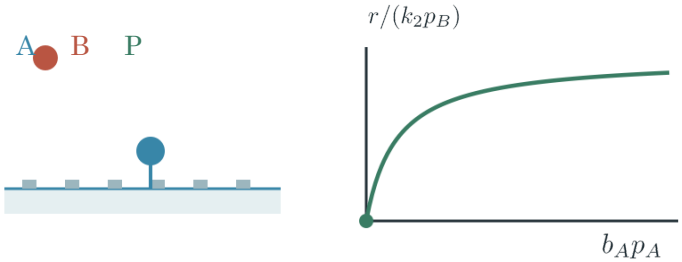
Langmuir-Hinshelwood rate law

The parameters $b_A b_B$ and the rate constant k_2 are all temperature dependent, so the overall temperature dependence of the rate may be strongly non-Arrhenius.

The $r-p_A$ plot has a maximum value if p_B is fixed (that is, $p_B \gg p_A$).

The Eley-Rideal Mechanism





The difference of this mechanism from LH mechanism is that the adsorbed A can only react with B in the gas phase but not with the adsorbed B.

Again, let's consider that the surface reaction is the rate-determining step (RDS), then the reaction rate is proportional to fractional coverage of A and partial pressure of B:

$$r = k_2 p_B \theta_A$$

Let's assume the Langmuir isotherm applies:

$$\theta_A = \frac{b_A p_A}{1+b_A p_A+b_B p_B} \qquad r = \frac{k_2 b_A p_A p_B}{1+b_A p_A+b_B p_B}$$

If the adsorption of B is weak or not happening:

$$r = \frac{k_2 b_A p_A p_B}{1+b_A p_A}$$

At low p_A, $b_A p_A \ll 1$, $r = k_2 b_A p_A p_B$

At high p_A, $b_A p_A \gg 1$, $r = k_2 p_B$

The r-p_A plot first increases with p_A and finally levels at a limiting value if p_B is fixed (that is, $p_B \gg p_A$).

Thus the r-p_A plot (with p_B is fixed) can be used to see a reaction follows LH or ER mechanism.

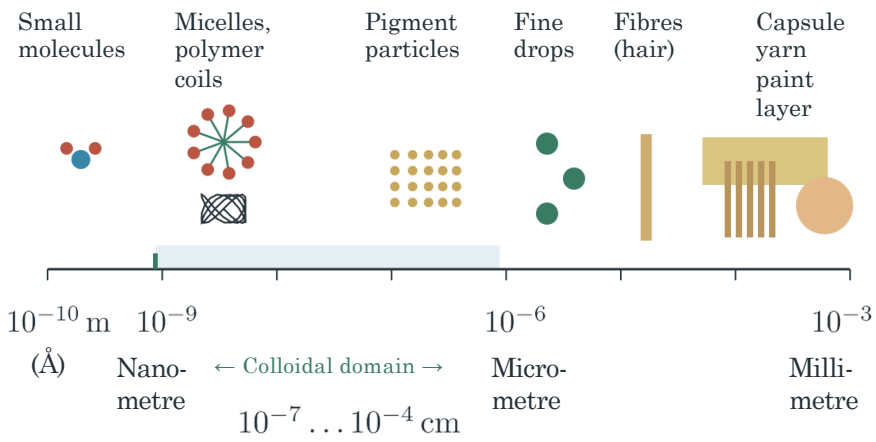
Colloid

'Colloidal' (胶体的) refers to a state of subdivision, implying that the molecules or polymolecular particles dispersed in a medium have at least in one direction a dimension roughly between 1 nm and 1 µm, or that in a system discontinuities are found at distances of that order. (IUPAC Gold Book)

A colloid, or disperse phase, is a dispersion of small particles of one material in another that does not settle out under gravity. In this context, 'small' means that one dimension at least is smaller than about 500 nm (about the wavelength of visible light). Many colloids are suspensions of nanoparticles (of diameter up to about 100 nm). In general, colloidal particles are aggregates of numerous atoms or molecules, but are usually too small to be seen with an optical microscope.

Key features of colloidal systems:

- multiple phase (多相)
- high dispersion (高度分散)
- thermodynamically unstable but kinetically non-labile (热力学不稳定但动力学稳定)



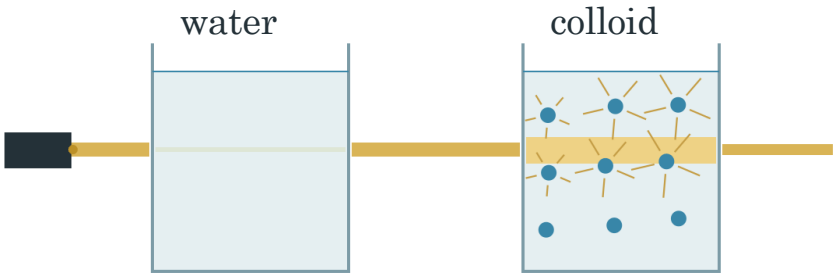
Dispersed Phase	Dispersion Medium	Name	Examples
Solid	Solid	Solid sol	Alloys, gems, coloured glass
Solid	Liquid	Sol	Paints, inks, egg white
Solid	Gas	Aerosol	Smoke, dust
Liquid	Solid	Gel	Curd, Cheese, jelly
Liquid	Liquid	Emulsion	Milk, butter, oil in water
Liquid	Gas	Aerosol	Mist, fog, clouds
Gas	Solid	Solid foam	Cake, bread
Gas	Liquid	Foam	Soap lather, Aerated water

Optical Properties

The Tyndall effect (丁达尔现象) can be described as the scattering of light by particles in a colloid or in a very fine suspension.

The Tyndall effect occurs from particles roughly the same size as the wavelength of light. When there is the Tyndall effect, light with longer wavelengths is transmitted, while light with shorter wavelengths is reflected via scattering. This effect can be observed when light-scattering particulate matter disperses in otherwise light-transmitting medium in which the diameter of an individual particle can be found in the range of roughly 40 – 900 nm, which is somewhat below or near the wavelengths of visible light.

The intensity of the scattered light is inversely proportional to the fourth power of the wavelength.



$$I = \frac{24\pi^3 \widehat{N} V^2}{\lambda^4} \left(\frac{n^2 - n_0^2}{n^2 + n_0^2} \right)^2 I_0$$

Rayleigh scattering formula

When light hits small particles, the light scatters in all directions (Rayleigh scattering) as long as the particles are small compared to the wavelength (below 250 nm). Dynamic light scattering (DLS) (动态激光散射) is a technique in physics that can be used to determine the size distribution profile of small particles in suspension or polymers in solution.

Dynamic Properties

The colloidal particles move randomly in a zigzag motion in all directions. This type of zigzag motion of colloidal particles is called Brownian movement. It is caused due to the collision of the colloidal particles with the molecules of the dispersion medium.

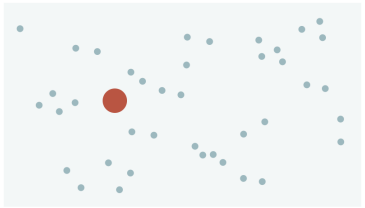
Einstein predicted that, just like a molecule in solution, such a Brownian particle would diffuse according to a simple equation:

$$\widehat{x}^2 = \frac{RT}{L}\frac{t}{3\pi\eta r} = 2Dt \Rightarrow \quad D = \frac{RT}{L}\frac{1}{6\pi\eta r} = \frac{kT}{6\pi\eta r}$$

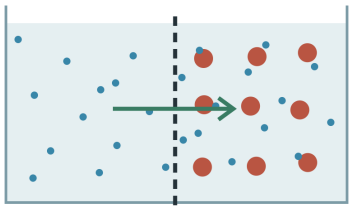
$\widehat{x}^2$: root mean square displacement

D : diffusion coefficient

Brownian motion



Osmosis



Assuming the colloidal particles are spheres, the particle size of the colloids (as well as the weight per colloid, supposed the density is known) can be known if the diffusion coefficient is determined:

$$r = \frac{kT}{6\pi\eta D} \qquad m = \tfrac{4}{3}\pi r^3\rho$$

r is the hydrodynamic radius

$$m = \frac{\rho}{162\pi^2}\left(\frac{kT}{\eta D}\right)$$

The r and m are averaged values if the colloid is not monodispersed.

Osmosis is a specific type of diffusion: it is the passage of water from a region of high water concentration through a semi-permeable membrane to a region of low water concentration. The osmosis pressure of a colloidal system can be calculated by:

$$\Pi = \frac{n}{V}RT$$

n: the amount of colloidal molecules/particles in a volume of V

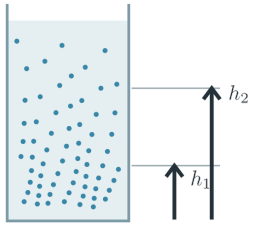
Dynamic Properties

Sedimentation (沉降) of colloidal particles occurs due to an external force, such as gravity or centrifugal force. The colloidal particles are said to be in sedimentation equilibrium (沉降平衡) if the rate of sedimentation is equal to that of Brownian motion.

Under gravity, the downward force responsible for sedimentation can be written as:

$$f_{\text{sedi}} = \frac{4}{3}\pi r^3(\rho - \rho_0)g$$

The distribution of particles follows the Boltzmann distribution and can be calculated as:



$$\frac{\widehat{N}_2}{\widehat{N}_1} = \exp\left[-\frac{3}{4}\pi r^3 \frac{(\rho-\rho_0)gL(h_2-h_1)}{RT}\right]$$

图 11-45 沉降平衡时浓度梯度稳定

When the sedimentation proceeds with a speed of v, the sedimentation force is balanced with the viscous drag, which equals to

$$F_{\text{viscous}} = 6\pi\eta r \cdot v$$

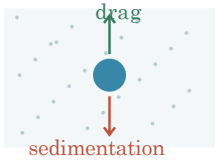
according to the Stokes' law. Thus, we have:

$$\frac{4}{3}\pi r^3(\rho - \rho_0)g = 6\pi\eta r \cdot v$$

$$v = \frac{2r^2}{9\eta}(\rho - \rho_0)g$$

The sedimentation speed is dependent on the second power of particle size.

The sedimentation speed can be tuned by adjusting the density difference or the viscosity.



The particle size or the mass can be known if the sedimentation speed can be determined.

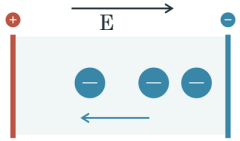
Electrical Properties

Because the colloidal particles are charged, a colloidal system exhibits electrokinetic phenomena (电动现象).

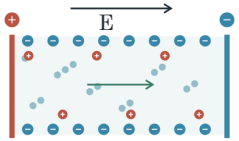
The unidirectional migration of the colloidal particles under the influence of an electric field is called electrophoresis (电泳). The potential difference developed between the lower layer and the upper layer in sedimentation is known as sedimentation potentials (沉降电势).

The migration of dispersion medium under the influence of an applied electric field (in which electrophoresis is prevented) in presence of a semipermeable membrane is known as electroosmosis (电渗). The potential developed when a colloidal solution is forced to pass through a porous membrane or capillary is known as streaming potential (流动电势).

Electrophoresis (电泳)



Electroosmosis (电渗)



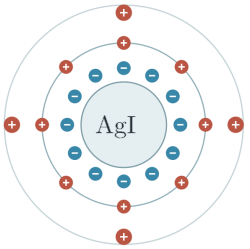
The reasons for the colloids being charged:

Surface adsorption (表面吸附) of ions

Fajans rule (Fajans规则): For inorganic sols, the colloids preferentially adsorb ions comprising itself.

Dissociation (电离) of surface molecules

Isoelectric point (等电点): The pH of a system at which the net charge of the colloids becomes zero.



胶核 AgI
吸附离子 I^-
紧密层 K^+
扩散层 K^+

胶粒 + 扩散层 = 胶团

$$[(AgI)_m \cdot n\,I^- \cdot (n-x)\,K^+]^{x-} \cdot x\,K^+$$

图 11-49 AgI 胶体粒子的结构

Electrical Properties

As the colloidal particles are charged, an electrical potential difference develops from the particle surface to the bulk dispersion medium:

Surface potential refers to the potential difference between the outer surface of adsorbed ions and the bulk solution.

The Stern layer moves constantly with the particle, but the diffuse layer does not. So the boundary between the two layers is called as slipping plane.

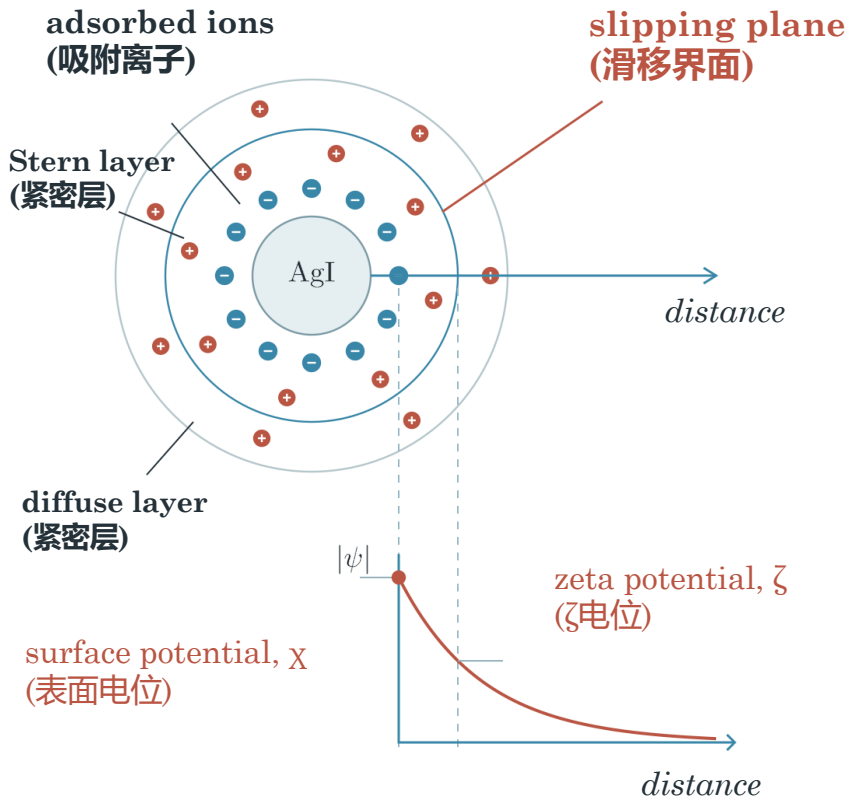
Zeta potential (ζ potential) refers to the potential difference between the slipping plane and the bulk solution.

Zeta potential signifies the charge a colloidal particle carries as well as the sign of the charge.

Zeta potential determines thickness of the diffuse layer as well as the speed of electrophoresis.

$$v = \frac{\varepsilon E \zeta}{K \eta}$$

ε: permittivity η: viscosity
K: a constant related to particle morphology



Colloidal Stability

Colloidal systems are thermodynamically unstable but kinetically non-labile, that is, they are not stable thermodynamically. The instability of a colloidal system exhibits in two ways:

Ostwald ripening (Ostwald老化) refers to a phenomenon that small crystals or sol particles first dissolve and then redeposit onto larger crystals or sol particles. This is a thermodynamically-driven spontaneous process because larger particles are more energetically favored than smaller particles. The rate of ripening is governed by diffusion.

Coagulation (聚沉) refers to a process of aggregation or accumulation of colloidal particles to settle down as a precipitate. The rate of coagulation is primarily affected by the electrical properties of the colloidal particles. Coagulate (聚沉剂) is a kind of substances, typically electrolytes, that can prompt coagulation.

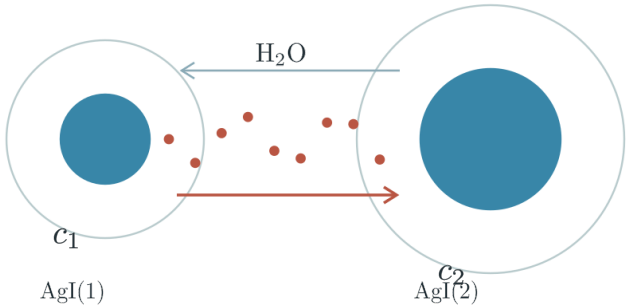
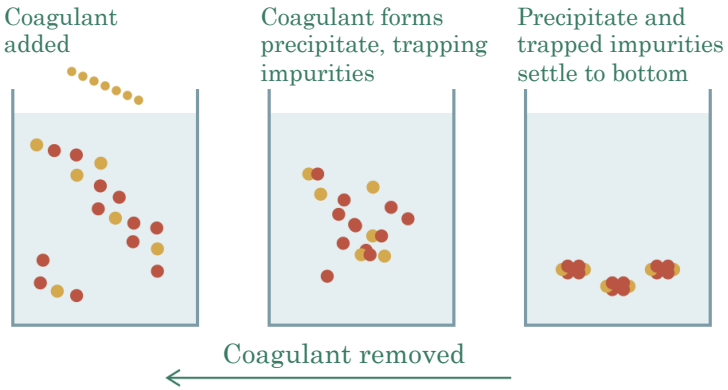


图 11-52 AgI 溶胶的老化

Aging is irreversible, as the large particles would not become redispersed.



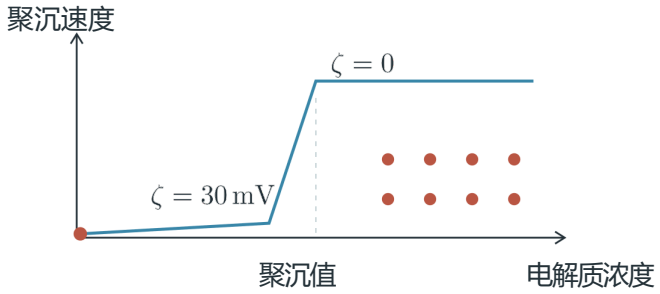
Coagulation can be reversible in certain conditions.

Colloidal Stability

A major source of kinetic non-lability of colloids is the existence of an electric charge on the surfaces of the particles. The addition of electrolytes, some of which can be adsorbed by the colloidal particles, can affect the colloidal stability. A electrolyte can either prompt or weaken the stability depending on whether it increases or decreases the ζ potential of the colloids.

The concentration of a coagulate, at which the coagulation process starts to take place apparently, is called critical coagulation value (CCC, 临界聚沉值).

The rate of coagulation reaches a maximum when ζ potential becomes zero, that is, at isoelectric point.



The counterions plays a key role in the coagulation of colloidal particles. The Schulze-Hardy rule suggests that the CCC of a coagulate decreases as the inverse sixth power of the counterion valence.

表 11-7 几种电解质的聚沉值 $\mathrm{mol \cdot m^{-3}}$

As_2S_3 (负溶胶)	CCC	AgI (负溶胶)	CCC
LiCl	58	$LiNO_3$	165
NaCl	51	$NaNO_3$	140
KCl	49.5	KNO_3	136
KNO_3	50	$RbNO_3$	126
$CaCl_2$	0.65	$Ca(NO_3)_2$	2.40
$MgCl_2$	0.72	$Mg(NO_3)_2$	2.60
$MgSO_4$	0.81	$Pb(NO_3)_2$	2.43
$AlCl_3$	0.093	$Al(NO_3)_3$	0.067
½ $Al_2(SO_4)_3$	0.096	$La(NO_3)_3$	0.069
$Al(NO_2)_3$	0.095	$Ce(NO_3)_3$	0.069

Lyotropic series (感胶离子序): For the ions of the same valence, the smaller the hydration sphere of the ion, the higher is its function of coagulation.

$$H^+ > Cs^+ > Rb^+ > NH_4^+ > K^+ > Na^+ > Li^+$$

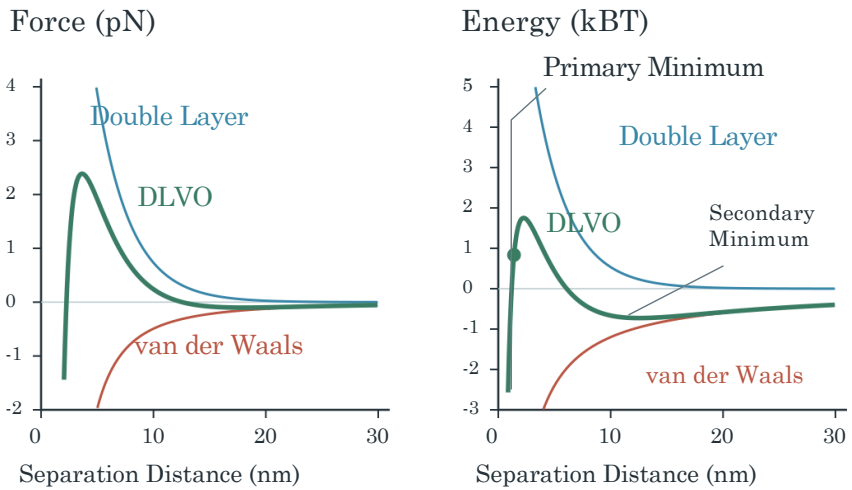
$$F^+ > Cl^+ > Br^+ > NO_3^- > I^+$$

The DLVO Theory

Derjaguin, Landau, Verwey, and Overbeek (DLVO) developed a theory of colloidal stability, which currently represents the cornerstone of our understanding of interactions between colloidal particles and their aggregation behavior.

DLVO theory assumes that the free energy per unit area can be well approximated by two additive contributions, namely van der Waals and electrical double layer interactions. The total potential energy is described as the sum of the attraction potential and the repulsion potential.

At very close distances, the combination of these forces results in a deep attractive well, which is referred to as the primary minimum. At larger distances, the energy profile goes through a maximum, or energy barrier, and subsequently passes through a shallow minimum, which is referred to as the secondary minimum.

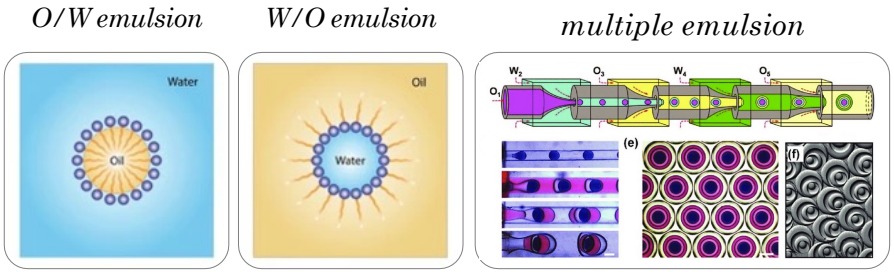


$$U_{\mathrm{DLVO}} = U_{\mathrm{vdW}} + U_{\mathrm{EDL}}$$

At primary minimum, attractive forces overpower the repulsive forces at low molecular distances. Particles coagulate and this process is not reversible. However, when the maximum energy barrier is too high to overcome, the colloid particles may stay in the secondary minimum, where particles are held together but more weakly than in the primary minimum. Particles form weak attractions but are easily redispersed. Thus, the adhesion at secondary minimum can be reversible.

Emulsions

- An ***emulsion*** (乳液) is a mixture of two or more liquids that are normally immiscible (不互溶的) owing to liquid-liquid phase separation.
- Two liquids can form different types of emulsions. As an example, oil and water can form, first, an oil-in-water emulsion (*O/W emulsion*), in which the oil is the dispersed phase, and water is the continuous phase. Second, they can form a water-in-oil emulsion (*W/O emulsion*), in which water is the dispersed phase and oil is the continuous phase. *Multiple emulsions* are also possible, including a "water-in-oil-in-water" emulsion and an "oil-in-water-in-oil" emulsion.
- An ***emulsifier*** (乳化剂) is an additive which helps to form and stabilize emulsions. Emulsifiers belong to the surfactants, usually with a grease-loving (*hydrophobic*, 疏水的) and a water-loving (*hydrophilic*, 亲水的) part, which can nest around boundary layers between aqueous and greasy parts. *Amphiphilic* (双亲性的) nanoparticles can also serve as surfactants/emulsifiers.



Emulsions are *thermodynamically unstable*, which means that phase separation would eventually happen. Two key factors determine the *kinetical non-liability* (动力学稳定性) of emulsions:

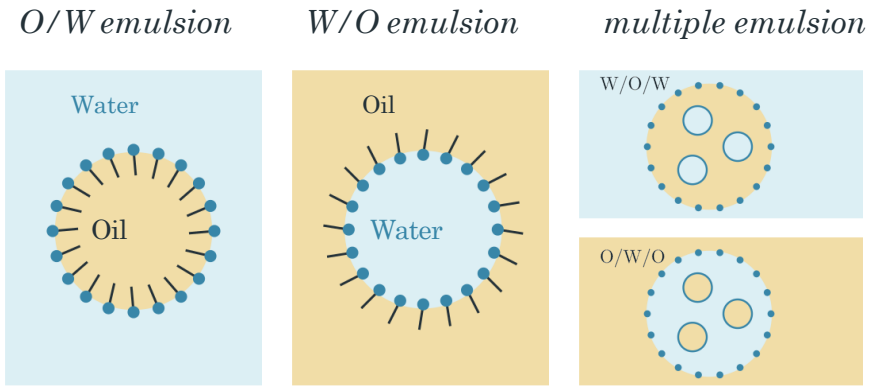
- Surface tension: In general, reducing the surface tension can help stabilize an emulsion.
- Strength of interfacial film: A stronger interfacial film is beneficial for emulsion stabilization.

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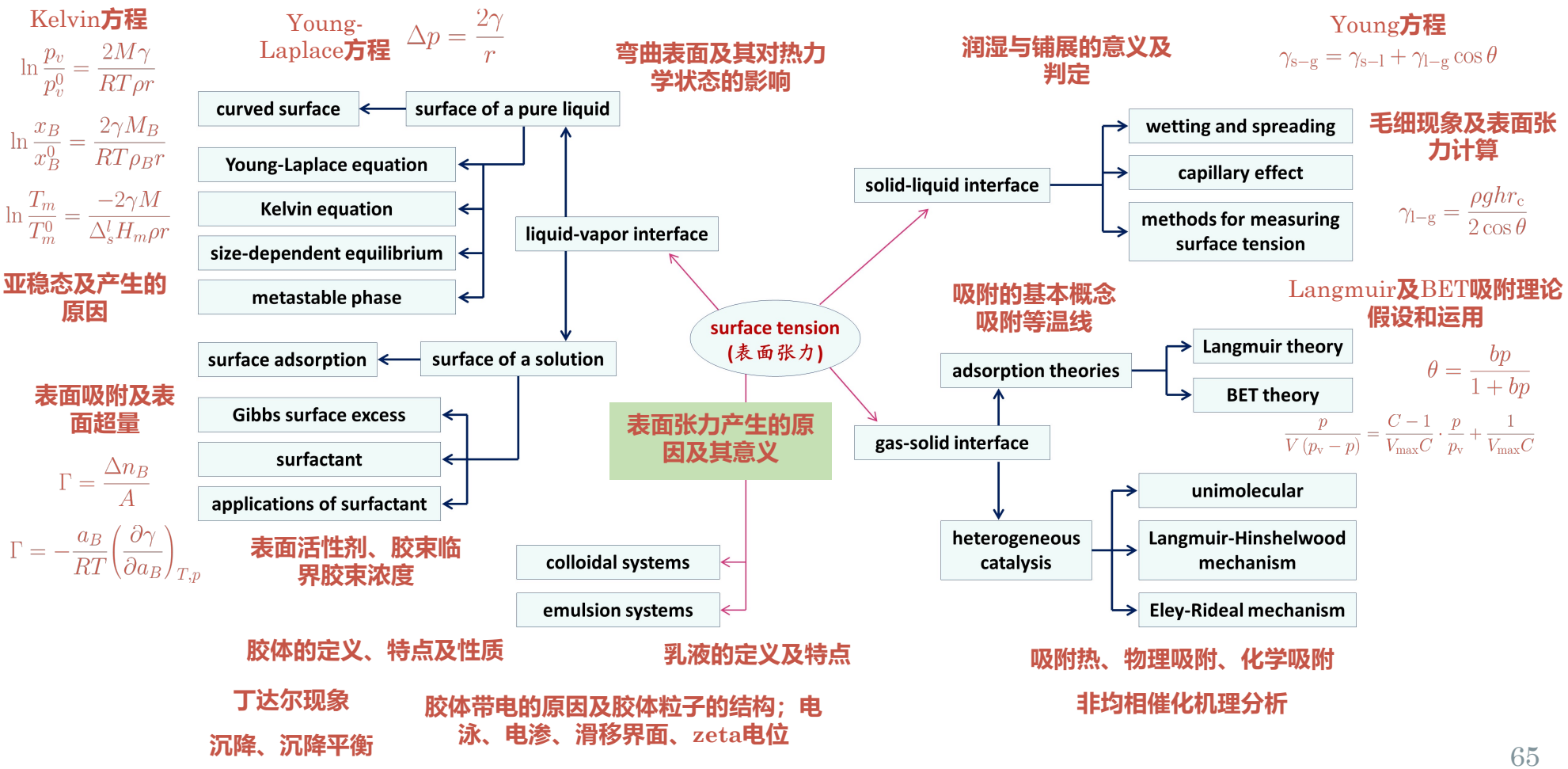


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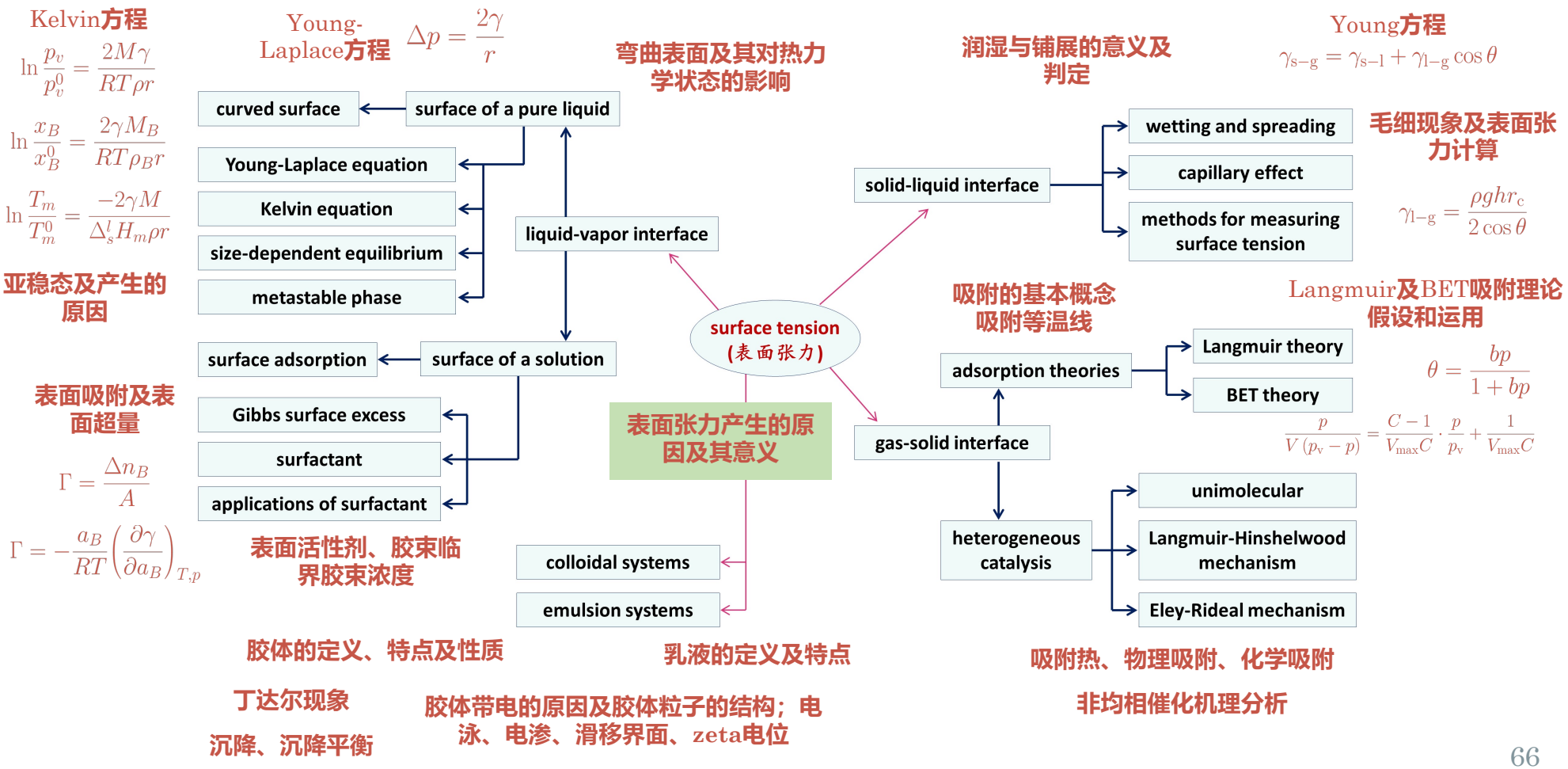
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Strength of interfacial film: A stronger interfacial film is beneficial for emulsion stabilization.

Summary


Kelvin方程
$\ln\frac{p_v}{p_v^0}=\frac{2M\gamma}{RT\rho r}$
$\ln\frac{x_B}{x_B^0}=\frac{2\gamma M_B}{RT\rho_B r}$
$\ln\frac{T_m}{T_m^0}=\frac{-2\gamma M}{\Delta_s^l H_m \rho r}$
亚稳态及产生的原因
Young-Laplace方程
$\Delta p=\frac{2\gamma}{r}$
弯曲表面及其对热力学状态的影响
curved surface
surface of a pure liquid
Young-Laplace equation
Kelvin equation
size-dependent equilibrium
metastable phase
liquid-vapor interface
surface adsorption
surface of a solution
Gibbs surface excess
surfactant
applications of surfactant
表面吸附及表面超量
$\Gamma=\frac{\Delta n_B}{A}$
$\Gamma=-\frac{a_B}{RT}\left(\frac{\partial\gamma}{\partial a_B}\right)_{T,p}$
表面活性剂、胶束临界胶束浓度
surface tension
(表面张力)
表面张力产生的原因及其意义
colloidal systems
emulsion systems
胶体的定义、特点及性质
乳液的定义及特点
丁达尔现象
沉降、沉降平衡
胶体带电的原因及胶体粒子的结构；电泳、电渗、滑移界面、zeta电位
润湿与铺展的意义及判定
Young方程
$\gamma_{s-g}=\gamma_{s-l}+\gamma_{l-g}\cos\theta$
solid-liquid interface
wetting and spreading
capillary effect
methods for measuring surface tension
毛细现象及表面张力计算
$\gamma_{l-g}=\frac{\rho g h r_c}{2\cos\theta}$
吸附的基本概念
吸附等温线
Langmuir及BET吸附理论假设和运用
adsorption theories
Langmuir theory
BET theory
$\theta=\frac{bp}{1+bp}$
$\frac{p}{V(p_v-p)}=\frac{C-1}{V_{max}C}\cdot\frac{p}{p_v}+\frac{1}{V_{max}C}$
gas-solid interface
heterogeneous catalysis
unimolecular
Langmuir-Hinshelwood mechanism
Eley-Rideal mechanism
吸附热、物理吸附、化学吸附
非均相催化机理分析

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吸附热、物理吸附、化学吸附
非均相催化机理分析

Aim

- ☑ You should ***understand*** the concept of surface tension and know how to use it to explain surface/interface phenomena.
- ☑ You should ***know*** how to use the Young-Laplace equation and the Kelvin equation to calculate equilibrium problems.
- ☑ You should ***understand*** the capillary effect and be able to judge whether a liquid wet or spread on a solid surface.
- ☑ You should ***know*** the methods to determine surface tension.
- ☑ You should ***know*** the definition of surfactants and their properties.
- ☑ You should ***master*** the Langmuir theory and the BET theory and can ***use*** them to calculate surface area and adsorption enthalpy.
- ☑ You should ***understand*** the mechanism of heterogenous catalysis.
- ☑ You should ***know*** the definition and properties of colloids.
- ☑ You should ***know*** the definition and properties of emulsions.

Check whether you have accomplished the above goals!